

Lawrence Berkeley National Laboratory

LBL Publications

Title

The water use of data center workloads: A review and assessment of key determinants

Permalink

<https://escholarship.org/uc/item/1vx545q7>

Journal

Resources Conservation and Recycling, 219

ISSN

0921-3449

Authors

Lei, Nuoa

Lu, Jun

Shehabi, Arman

et al.

Publication Date

2025-06-01

DOI

10.1016/j.resconrec.2025.108310

Copyright Information

This work is made available under the terms of a Creative Commons Attribution License, available at <https://creativecommons.org/licenses/by/4.0/>

Peer reviewed

The Water Use of Data Center Workloads: A Review and Assessment of Key Determinants

Nuoa Lei ¹, Jun Lu ², Arman Shehabi ¹, and Eric Masanet ^{3,4}

¹Energy Technologies Area, Lawrence Berkeley National Laboratory, Berkeley, CA, USA

²Division of Epidemiology and Biostatistics, University of Illinois, Chicago, IL, USA

³Bren School of Environmental Science and Management, University of California, Santa Barbara, CA, USA

⁴Department of Mechanical Engineering, University of California, Santa Barbara

Abstract

The global importance of data center water use is increasing with the rapid growth of digitalization and artificial intelligence. This study analyzes the factors influencing workload-level water use, measured in liters consumed per workload, to guide water-saving strategies in data centers. Our findings reveal workload-level water use variations exceeding 10,000-fold, driven by over 1,000-fold differences in water consumption per kilowatt hour of server electricity consumed and approximately 10-fold differences in server workload efficiency. Key determinants are ranked as server efficiency, electrical grid water consumption factors, server utilization, cooling system type, infrastructure efficiency, climate zone, inactive server percentage, and server refresh cycle. Notably, there is no single recipe for minimizing water use; instead, optimal outcomes depend on tailored combinations of these factors. This analysis addresses critical knowledge gaps by identifying the determinants of data center water use and exploring their achievable minima under diverse site-specific constraints.

Keywords

Data centers; Water; Water efficiency; Water-energy nexus; Water usage effectiveness.

1. Introduction

Increasing demand for digital services has led to an increased reliance on data centers—the backbone of the internet which provides the necessary functions of data processing, communication, and storage (Shehabi et al., 2024, 2016). However, the operation of data centers requires significant amounts of electricity and water (Masanet et al., 2020; Mytton, 2021). While data center electricity use has been the subject of much investigation (Andrae and Edler, 2015; Belkhir and Elmeligi, 2018; Jones, 2018; Lei and Masanet, 2021; Masanet et al., 2020), data center water consumption is a less studied topic that is becoming increasingly important in the context of water scarcity and freshwater conservation, especially at local levels (Lei and Masanet, 2022; Mytton, 2021; Siddik et al., 2021). Many data centers can consume a significant amount of water directly for cooling and space conditioning (Lei and Masanet, 2022). For example, Google's data centers in The Dalles consumed 355 million gallons of water in 2021, representing 29% of the city's total water consumption, which has raised concerns about local water stress, particularly in light of the recent drought conditions in this region (Oregon Tech, 2022). Additionally, data centers consume water indirectly through the water used for electricity generation (Mytton, 2021; Shehabi et al., 2016; Siddik et al., 2024, 2021). This type of water use can be significant at the local or national level depending on the power generation technologies, power mixes, and electricity usage of the data centers. For example, some very large data centers have power capacities of hundreds of megawatts and the generation of 1 kilowatt-hour of thermoelectricity or hydropower could consume 0.18-2.0 liters or 0.67-1194 liters of water (Lee et al., 2018), respectively. This demonstrates a significant amount of data center indirect water use and further emphasizes the importance of considering data center water use going forward.

To align with the growing importance of water conservation and sustainability in data centers, operators are beginning to track, manage, and report their water usage effectiveness (WUE) at the site level (WUE-site) and at the source level (WUE-source) (Facebook, 2014; Scaleway, 2021). WUE-site measures the efficiency of onsite water consumption at a data center facility, expressed as the ratio of the data center's total onsite water use to its information technology (IT) equipment electricity use. WUE-source quantifies water use efficiency at the source level by considering the total water use (including on-site and off-site

water) in relation to IT equipment electricity use (The Green Grid, 2011a). WUE-site and WUE-source are becoming more commonly-reported metrics within the data center industry due to their simplicity (Meta Sustainability, 2023). However, much like power utilization effectiveness (PUE) (Masanet et al., 2013), they do not consider the hardware efficiency or computing performance of a data center (Horner and Azevedo, 2016). Therefore, WUE-site and WUE-source cannot be used to compare the water use of data centers with different workload quantities and efficiencies.

In addition, systematic reporting and research related to data center water uses and their determinants remain limited in the literature, which poses challenges for designers, engineers, operators, investors, and policymakers in effectively understanding and addressing data center water usage issues.

- First, while reporting water use and efficiency metrics such as PUE and WUE-site is becoming more common in the industry, these metrics are primarily reported by cloud providers, with significant variation in scope, quality, and granularity (Masanet et al., 2024). Companies like Google, Meta, Microsoft, and Scaleway serve as role models for reporting standards (Google, 2024; Meta Sustainability, 2023; Microsoft, 2024; Scaleway, 2021), yet even their reports lack sufficient detail on water use determinants needed to inform effective water-saving strategies.
- Second, limited quantitative studies have explored the impact of cooling systems and economizers on WUE-site (Gozcu et al., 2017; Lei and Masanet, 2022; Sharma et al., 2009, 2008). Among these, (Lei and Masanet, 2022) offers the most comprehensive analysis, incorporating data center operating efficiency, climate conditions, and the combined effects of cooling systems and economizers on both PUE and WUE-site. However, this study does not account for the water that is indirectly used in electricity generation. Conversely, research from AB Siddik (Siddik et al., 2021) includes indirect water consumption but relies on summarized PUE statistics by data center size, derived from Masanet et al. (Masanet et al., 2020), which limits the ability to identify specific determinants of water use and develop targeted saving strategies.
- Third, another line of research conducted by Lawrence Berkeley National Laboratory has focused on estimating data center water use at the regional level (Shehabi et al., 2024, 2016). Their analysis

estimated that U.S. data centers directly consumed approximately 21.2 billion liters of water in 2014 and 66 billion liters in 2023. However, these studies disaggregate water use by space type and lack fine-grained analyses of the factors influencing water use, hindering the identification of tailored strategies to improve water efficiency.

- Finally, server workload efficiency, a key factor influencing water use per digital service, is often overlooked by sustainability practitioners. Workload efficiency is affected by variables such as server utilization, power efficiency, refresh rates, and inactive server percentages (Masanet et al., 2020; Shehabi et al., 2016). Addressing these factors is essential for a comprehensive understanding of data center water use per workload, which should account for both WUE-site and WUE-source determinants alongside the factors influencing server workload efficiency.

In this study, we aim to fill above knowledge gaps through a comprehensive analysis of overall data center water use per workload (Fig. 1). To make a fair comparison and gain a deeper understanding of the water use of data centers at the workload hosting level, we integrated the workload throughput score (i.e., *ssj_ops*) from the SPECpower_ssj 2008 benchmark into our analysis (Standard Performance Evaluation Corporation, 2022). This allowed us to create a new metric: the overall data center water use per workload, expressed in liters per workload ($\frac{\text{liters}}{\text{ssj_ops}}$), which addresses the limitations of the WUE-site and WUE-source metrics (Horner and Azevedo, 2016). We follow this with a decomposition analysis of the defined overall data center water use per workload, analyzing the determinants of water use and their interrelationships through the examination of real-world situations. In the decomposition analysis, we examined three key aspects of data center water use per workload in step-by-step fashion to make the influence of key determinants clear: (1) the relationship between server workload efficiency and factors such as server utilization level, power efficiency, refresh cycles, and inactive server percentage; (2) the effect of cooling technology, climate, and infrastructure efficiency on PUE, WUE-site, and their relationships to each other as well as how these interact with server workload efficiency to further impact the relationship between

facility electricity and water use per workload; and (3) the determinants of WUE-source and how they interact with server workload efficiency to further influence the overall data center water use per workload. Our analysis confirmed that considering workload efficiency is crucial in comparing water use between data centers, and the measures identified in our analysis can be effective for optimizing data center water use per computing workload from different perspectives. Additionally, our work can be readily extended to other measures of data center workloads derived from different testing methods when sufficient data exist, such as MLPerf for artificial intelligence workloads (MLCommons, 2024), using the framework we present. This is an important step towards achieving improved water use efficiency in data centers.

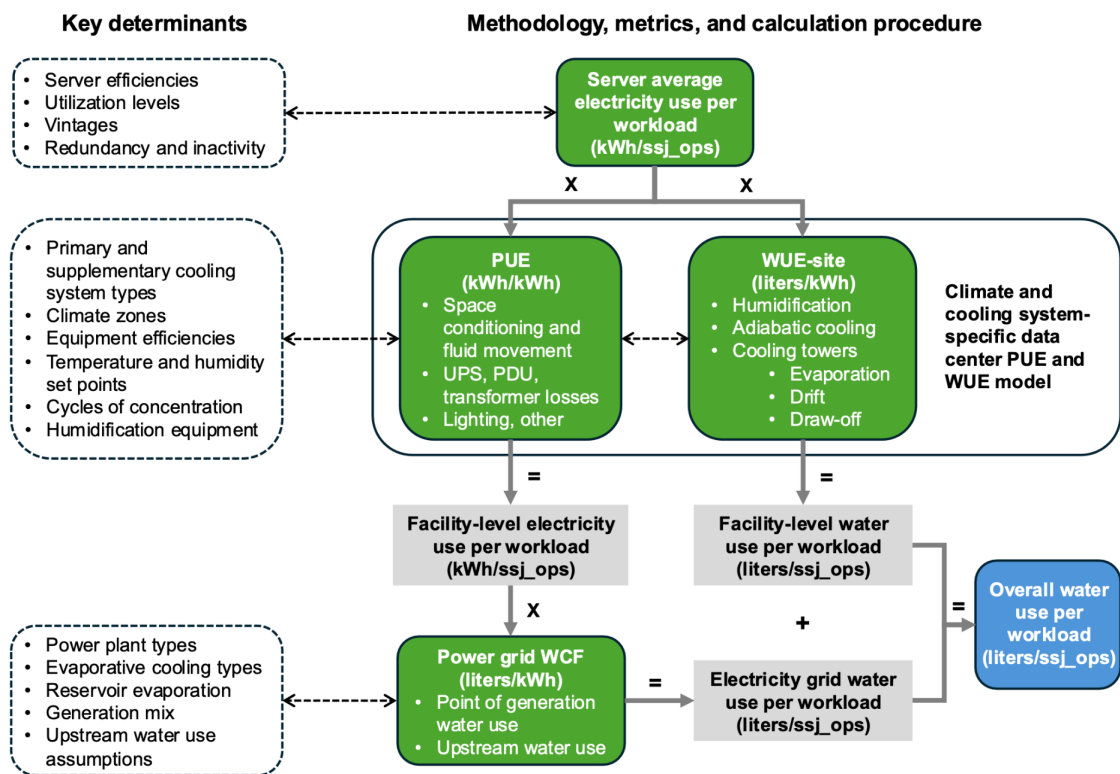


Fig. 1. Schematic of the overall water use per workload analysis approach and key systems determinants affecting the results. Determinants are depicted in dashed boxes linked to the variables (in green boxes) that they influence. The basic calculation procedure is indicated by grey arrows, with intermediate results highlighted in grey boxes.

2. Methods

2.1. Data center water use

The scope of this water use analysis includes both direct and indirect water consumption by data centers per workload. Direct water use encompasses water consumed at the data center for space humidification, direct adiabatic cooling of air, and/or heat rejection in cooling towers (including evaporated water, windage loss, and draw-off water). Indirect water use refers to water consumed through electricity generation, covering both upstream water use and water consumed at the point of power generation. Upstream water use includes water consumed in mining, extracting, cleaning, and processing of thermal power plant fuel inputs and in the manufacturing of renewable power equipment, such as solar photovoltaic (PV) panels and wind turbines. Point of generation water use refers to water consumed at the power plant itself due to the use of cooling towers, evaporation in cooling ponds and hydroelectric reservoirs, and periodic cleaning of solar PV panels.

Our decomposition analysis utilizes the equations outlined below, with the corresponding inputs and their sources summarized in Section S10 of the supporting information (SI). The overall data center water use per workload ($\frac{\text{liters}}{\text{ssj_ops}}$) is defined as Eq. (1).

$$W_{\text{overall}} = W_{\text{site}} + E_{\text{site}} \times WCF \quad (1)$$

where W_{site} is the facility-level water use per workload ($\frac{\text{liters}}{\text{ssj_ops}}$, Eq. (2)), E_{site} is the facility-level electricity use per workload ($\frac{\text{kWh}}{\text{ssj_ops}}$, Eq. (3)), and WCF is the electrical grid water consumption factor, defined as the amount of freshwater consumed (in liters) per kilowatt-hour ($\frac{\text{liters}}{\text{kWh}}$) of grid electricity generated (Lee et al., 2018).

$$W_{\text{site}} = WUE_{\text{site}} \times e_{\text{server}} \quad (2)$$

where WUE_{site} is the water usage effectiveness at the data center facility level ($\frac{\text{liters}}{\text{kWh}}$), e_{server} is the average server electricity use per workload (i.e., server workload efficiency) ($\frac{\text{kWh}}{\text{ssj_ops}}$, Eq. (5)).

$$E_{site} = PUE \times e_{server} \quad (3)$$

where PUE is the power usage effectiveness, defined as the ratio of a data center's total electricity use to its IT equipment electricity use ($\frac{kWh}{kWh}$) (The Green Grid, 2011b).

Substituting Eq. (2) and (3) into Eq. (1) gives Eq. (4):

$$W_{overall} = (WUE_{site} + PUE \times WCF) \times e_{server} = WUE_{source} \times e_{server} \quad (4)$$

where WUE_{source} is the source-based water usage effectiveness ($\frac{liters}{kWh}$) (The Green Grid, 2011a).

2.2. Direct server electricity use per workload

We focused on servers as the main component of IT loads because they are responsible for workload hosting, which is the main digital service associated with most data centers. However, future work could explore additional metrics for data center services, such as AI training and inference performance (Masanet et al., 2024), or installed storage capacity in storage focused data centers. The server electricity use per workload at the data center level was modeled as a function of average server utilization level, the vintage distribution of the installed server stock (or refresh cycle), average server efficiency, and the fraction of inactive servers (including comatose servers and redundant servers operating at idle state) within the installed server stock (Lei, 2022; Lei and Masanet, 2021; Shehabi et al., 2016), expressed as Eq. (5).

$$e_{server} = 3.6 \times \frac{\bar{P}}{\bar{Th}} = 3.6 \times \frac{\varphi \bar{P}_i + (1-\varphi) \bar{P}_a}{(1-\varphi) \bar{P}_a \overline{Perf}} \quad (5)$$

where $\bar{P}$ is the weighted-average power use of the servers (W), and $\bar{Th}$ is the weighted-average server throughput ($\frac{ssj_ops}{s}$). $\bar{P}_i$ is the average power use of idle servers (W , Eq. (6)), $\bar{P}_a$ is the average power use of active servers (W , Eq. (7)), $\overline{Perf}$ is the average server power-performance (ssj_ops/W , Eq. (8)), and φ is the fraction of inactive servers in the stock.

$$\bar{P}_i = \sum_y \phi_y \times P(u = 0, y) \quad (6)$$

$$\bar{P}_a = \sum_y \phi_y \times P(u, y) \quad (7)$$

$$\overline{Perf} = \sum_y \phi_y \times Perf(u, y) \quad (8)$$

where ϕ_y is the server vintage distribution, represented by the percentage of installed servers of year y in the server stock. $P(\cdot)$ and $Perf(\cdot)$ are, respectively, the server power and power-performance model, which are formulated as functions of the server utilization level (u) and the server installed year (y) using the local linear kernel regression approach based on the SPECpower_ssj 2008 benchmark database (Standard Performance Evaluation Corporation, 2022). Due to variability in server power-performance each year y , we analyzed $P(\cdot)$ and $Perf(\cdot)$ at different power efficiency quantiles for the pool of servers reported by the SPECpower_ssj 2008 database. Further details are available in SI Sections S1 and S2.

2.3. Climate and technology-specific PUE and WUE-site

PUE and WUE-site values are two generally-accepted metrics to report the energy and water use of data center operations (The Green Grid, 2011b, 2011a). As demonstrated in previous studies, the PUE and WUE-site values of data centers are simultaneously influenced by cooling system technologies, climate conditions, and operating efficiency practices (Lei and Masanet, 2022, 2020). However, they are also sporadically reported by operators and are poorly understood by sustainability analysts (Mytton, 2021). Given their importance and sometimes competing effects in determining overall data center water use (Eq. (4)), we conducted an in-depth review and simulations of PUE and WUE-site across different cooling systems and climate scenarios (SI Section S3). In these simulations, the PUE results account for all heat generation sources and power-consuming devices, including power used by IT equipment, lighting, fans, pumps, and losses from power transmission and distribution (Lei and Masanet, 2020). Meanwhile, WUE-site results consider water consumption associated with direct adiabatic cooling, humidification systems, and cooling tower operations (where applicable) (Lei and Masanet, 2022). Specifically, cooling tower water use includes water consumed for heat removal (sometimes referred to as greywater (Aldaya et al., 2012) in its functional role), windage loss, and draw-off water losses necessary to control chloride or mineral concentrations. Our aim was to clarify the interrelationships between PUE and WUE-site values and further illustrate how their determinants affect overall data center water use in the current study. To thoroughly quantify the impact of

the various determinants, we covered typical data center cooling technologies and operating conditions (Evans, 2004; Lei and Masanet, 2022, 2020), as well as major types of global climate zones specified by American Society of Heating, Refrigerating and Air-Conditioning Engineers (ASHRAE) (ASHRAE, 2013). In total, we considered ten cooling technologies including (1) air-cooled chillers with no economizer, (2) airside economizers using air-cooled chillers as supplemental cooling, (3) airside economizers using water-cooled chillers as supplement cooling, (4) airside economizers and direct adiabatic cooling using air-cooled chillers as supplemental cooling, (5) airside economizers and direct adiabatic cooling using water-cooled chillers as supplemental cooling, (6) direct expansion systems, (7) water-cooled chillers with no economizer, (8) waterside economizers using water-cooled chillers as supplemental cooling, (9) IT liquid cooling using waterside economizers and using water-cooled chillers as supplemental cooling, (10) IT liquid cooling using dry coolers (a type of air-cooled heat exchanger) supporting adiabatic assist and using air-cooled chillers as supplemental cooling. Note that in these cases, supplemental cooling refers to cooling systems that are activated only when the primary cooling method (e.g., economizers, dry coolers, or adiabatic cooling) cannot maintain the desired temperature setpoints, which typically occurs during extreme weather conditions when free cooling methods alone cannot adequately dissipate heat. Additionally, we considered data center operations under nineteen different climate zones including extremely hot humid (0A), extremely hot dry (0B), very hot humid (1A), very hot dry (1B), hot humid (2A), hot dry (2B), warm humid (3A), warm dry (3B), warm marine (3C), mixed humid (4A), mixed dry (4B), mixed marine (4C), cool humid (5A), cool dry (5B), cool marine (5C), cool humid (6A), cold dry (6B), very cold (7), subarctic/arctic (8). Collectively, our combinations of cooling technologies and climate zones provide comprehensive global coverage. Finally, we took different data center operating efficiency practices, which were documented in previous literature, into consideration (Lei and Masanet, 2022, 2021, 2020). These different efficiency practices are represented by different quantiles of PUE and WUE-site simulation values for data centers implementing a given cooling technology in a given climate zone. Further details are provided in Section S3 of the SI.

2.4. Electric power water consumption factors

To thoroughly explore the indirect water use of data centers and to quantify its contribution to overall water use per workload, we analyzed water consumption factors (WCFs) across various electricity generation sources. Specifically, we assessed how different power sources—including geothermal, hydro, biomass, oil, coal, natural gas, solar photovoltaics (PV), and wind—affect overall data center water use per workload. This analysis covers both upstream water uses (e.g., water used in fuel extraction, processing, and transportation), and water consumed during power generation (Peer et al., 2019; Siddik et al., 2021), offering a more comprehensive assessment of water usage across the electricity generation life cycle (SI Section S4). However, it should be noted that the WCFs used in this study are based on generation-weighted U.S. averages. Since WCFs can vary significantly by region, power plant cooling technology, prime mover technology, and allocation methods (Grubert and Sanders, 2018; Lee et al., 2018), future research could investigate the impact of these factors to better align with the goal of accurately assessing their contribution to overall data center water use in non-U.S. regions.

3. Results

3.1. Determinants of direct server electricity use per workload

Our decomposition analysis starts at analyzing the factors that contribute to the server electricity use per data center workload. Fig. 2 shows the impact of various factors on server electricity use per workload, obtained using Eqs. (5) ~ (8) and assuming that the newest servers in the data center were purchased in 2023. The result shows that the lowest electricity use per workload is generally associated with efficient servers, high server utilization levels, fast refresh cycles, and low fractions of inactive servers. However, none of these factors alone is usually sufficient to justify low direct electricity use per workload, as indicated by how different combinations of these factors result in the same server electricity per workload (i.e., the same color) in Fig. 2. For example, a high-efficiency server pool (95th percentile) running at a low average utilization level and with many inactive servers can consume the same electricity per workload as a low-efficiency server pool (5th percentile) running at high utilization levels with few inactive servers (see points A and B, respectively). Server electricity use per workload is mainly affected by server efficiency. Efficient

1 servers consume less power at idle state, and typically have larger dynamic ranges and better power
2 proportionality (Barroso and Hölzle, 2007; Ryckbosch et al., 2011), leading to lower electricity use per
3 workload (SI Fig. S5.1). Increasing server utilization level is another strategy to reduce direct electricity
4 use per workload in data centers; however, the benefits diminish once server utilization exceeds around
5 40% (SI Fig. S5.2). Optimizing the server refresh cycle reduces electricity use per workload, and at all
6 utilization levels (SI Fig. S5.3), in a manner similar to using power efficient servers because servers become
7 more power efficient over time. Likewise, removing inactive servers from the data center can reduce total
8 electricity use per workload (SI Fig. S5.4), since idle servers consume power but are not performing any
9 computing tasks. However, newer generations of servers also have significantly lower idle power,
10 diminishing the effect of inactive servers over time.

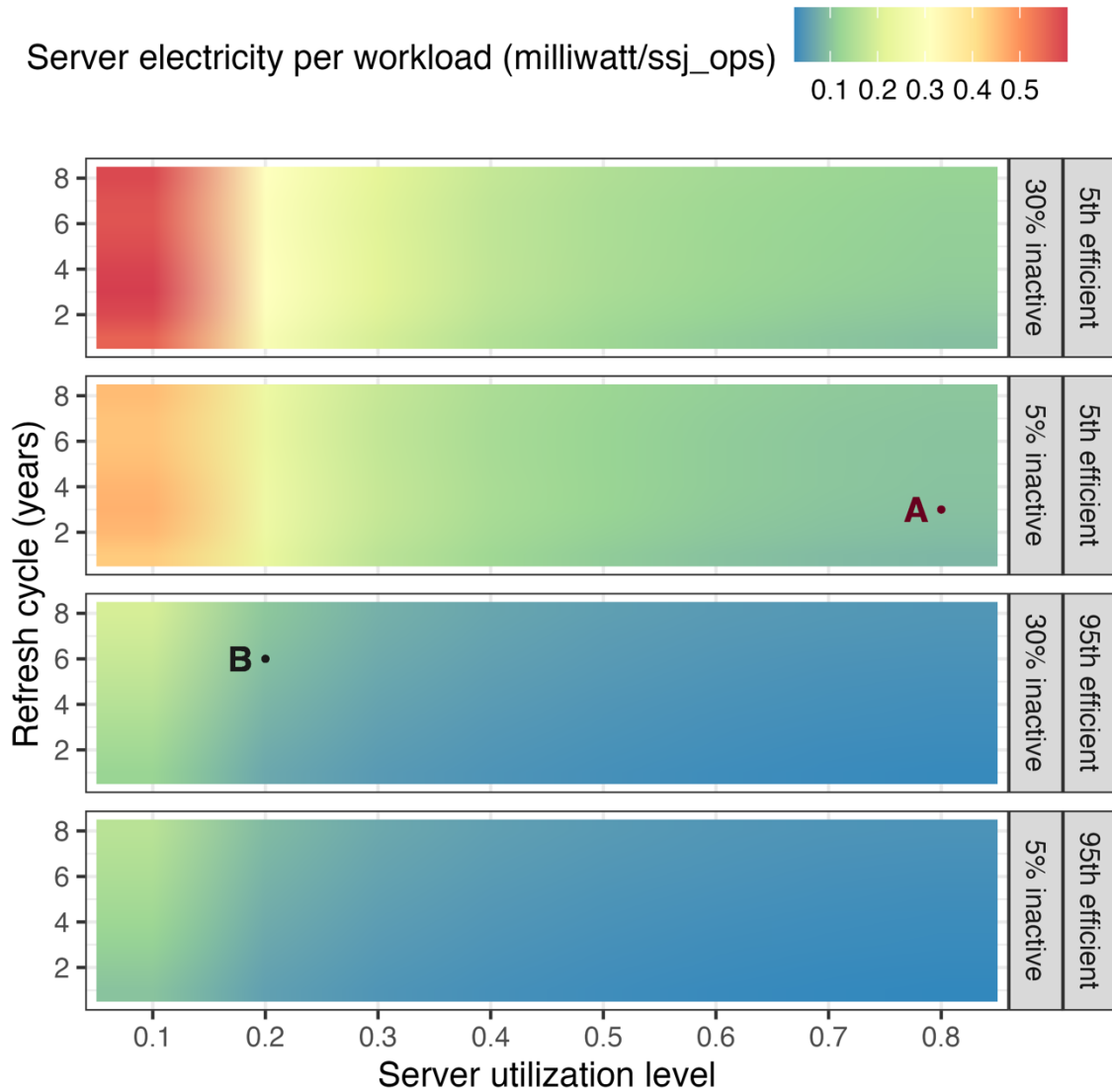


Fig. 2. Determinants of server electricity use per workload.

Notes: 5% inactive indicates 5% percent of the server pool is inactive servers; 30% inactive indicates 30% percent of the server pool is inactive servers; 5th efficient indicates the lowest 5% of server power efficiency, i.e., the 5th percentile; 95th efficient indicates the highest 5% of server power efficiency, i.e., the 95th percentile.

1

2 3.2. Determinants of facility-level water use per server workload

3 Next in our decomposition analysis, we analyzed facility-level water use per workload as expressed in Eq.

4 (2). Fig. 3 shows the assessed relationships between WUE-site, server electricity use per workload, and

5 facility water use per workload. For a given server electricity use per workload, facility water use per

6 workload increases with the increase of WUE-site, whereas server electricity use per workload determines

the rate of this increase. Notably, low facility water use per workload is not necessarily a result of a low WUE-site value. This situation is indicated by points A and B in Fig. 3, in which both points have the same facility water use per workload but have vastly different values of WUE-site and server electricity use per workload. The interrelationships highlighted in Fig. 3 demonstrate the necessity of considering both WUE-site and server workload efficiency when considering the water efficiency of a given data center, as opposed to WUE-site alone.

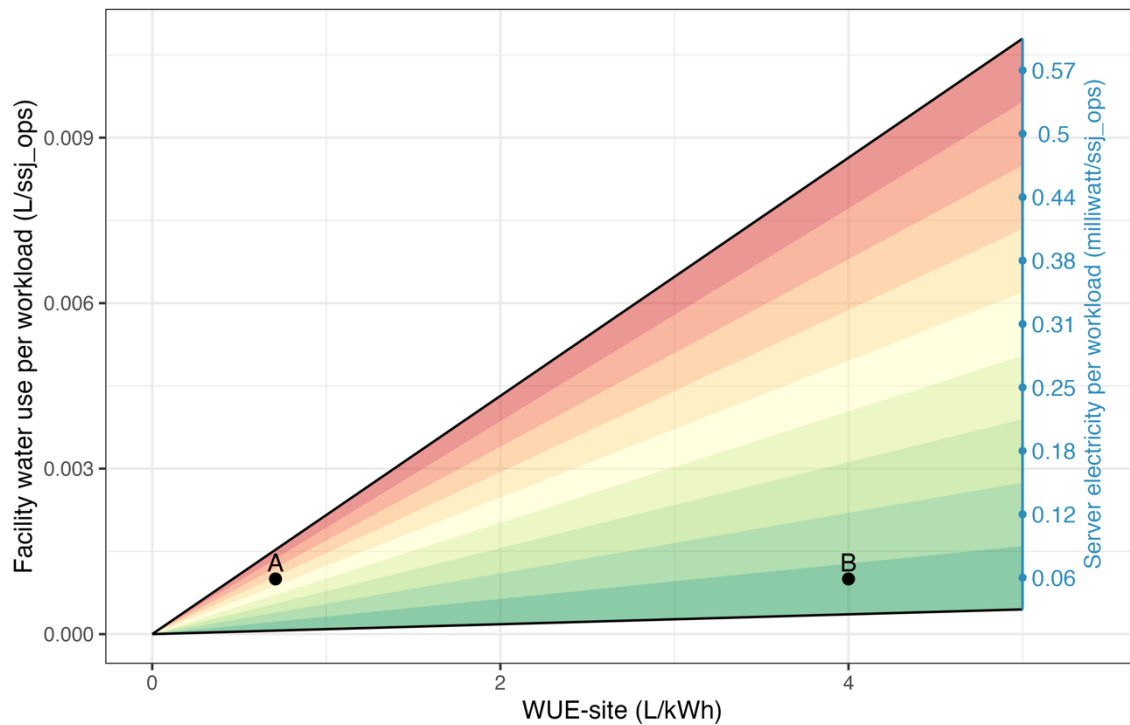


Fig. 3. The relationship between WUE-site, server electricity per workload, and facility-level water use per workload.

Note: the server electricity per workload axis values were uniformly sampled between the minimum and the maximum value in Fig. 2.

3.3. Relationships between PUE and WUE-site values

Fig. 4. shows the interrelationships between PUE and WUE-site values clustered by ten different cooling technologies. The colored regions depicted for each cooling technology represent the range of variations in the relationship between PUE and WUE-site attributable to the combined effects of climate and data center infrastructure operating efficiency (see SI Fig. S6.1, which depicts how a region was defined for one cooling

technology). It can be clearly observed that PUE and WUE-site values vary significantly by cooling technologies. Data centers that employ water-cooled chillers (as primary or supplemental cooling) have relatively higher WUE-site values due to large amounts of water consumed by cooling towers, particularly from evaporation. Conversely, data centers using direct-expansion systems or air-cooled chillers (as primary or supplemental cooling) have relatively lower WUE-site values, but typically have higher PUE values due to lower chiller coefficient of performance (COP) values (Lei and Masanet, 2022). Additionally, data centers that employ liquid cooling coupled with dry coolers, economizers, or adiabatic cooling systems can reduce both PUE and WUE-site values simultaneously, with the greatest electricity and water savings achieved for data centers using airside economizers and direct adiabatic cooling. Notably, data centers adopting IT liquid cooling systems can further reduce their PUE and WUE-site values owing to enhanced heat transfer efficiency, elevated coolant/facility water temperature setpoints, and the potential reduction or elimination of fan and chiller power (ASHRAE, 2014; Iyengar et al., 2012). Within each cooling technology cluster, we note a non-linear and positive relationship between PUE and WUE-site, where favorable climates and high operational efficiencies result in low PUE and WUE-site values, and hot and/or humid climates and low operational efficiencies lead to high PUE and WUE-site values (SI Fig. S6.1). Finally, we note that similar PUE and WUE-site values can be achieved with different combinations of cooling technology, climate, and efficiency, represented by overlaps between clusters. For example, points A and B have the same WUE-site value but PUE values differ by 2.0, while points C and D have the same PUE value but WUE-site values differ by 2.6. This situation again highlights the importance of system-wide optimization that simultaneously considers technology, climate, and efficiency in PUE and WUE-site reduction.

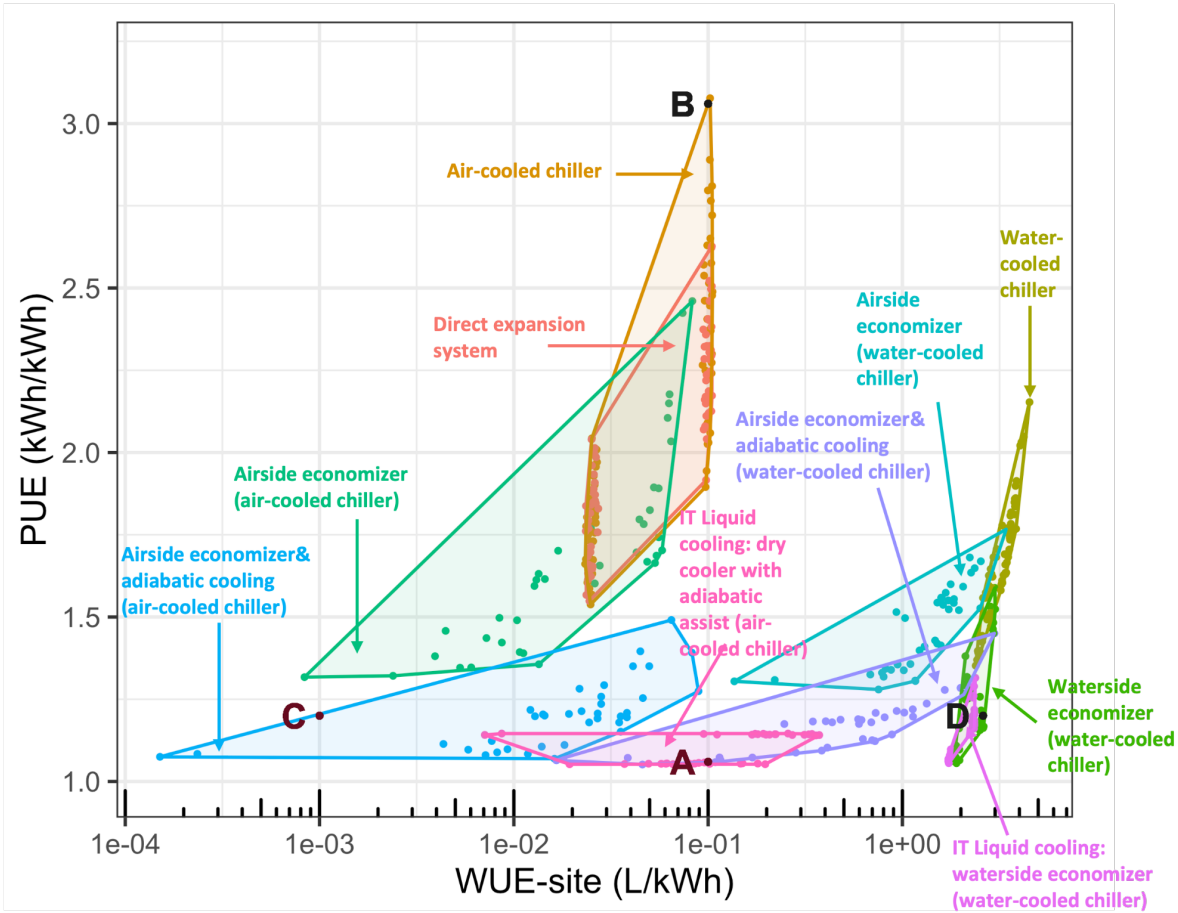


Fig. 4. Relationship between PUE, WUE-site, and cooling technology (x-axis values are logarithmically spaced).

Notes: See SI. Fig. S6.2 (where the x-axis is linearly spaced); Scatters within a given cooling technology consist of PUE and WUE-site values of different climate zones and data center operating efficiency practices (SI. Fig. S6.1); air-/water-cooled chillers in parentheses stand for supplemental cooling sources; PUE and WUE-site are yearly average values; IT liquid cooling includes scenarios of rear door heat exchangers, cold-plates, and immersion cooling (SI. Fig. S6.3).

1

2 3.4. Relationships between facility-level electricity use and water use per workload

3 As shown in Fig. 3, facility-level water use per workload is driven by WUE-site and average server
4 workload efficiency. Combining the regions in Figs. 2 and 3 (via Eqs. (2) and (3)) yields Fig. 5, which plots
5 the relationship between facility-level water use and facility-level electricity use on a per-workload basis.
6 In Fig. 5, large regions of possible facility-level electricity and water performance emerge due the combined
7 variations associated with climate, operational efficiencies, and server workload efficiencies.

1 Our analysis showed that the lowest data center facility-level electricity and water use per workload are
2 generally associated with the highest server workload efficiencies (SI Fig. S7.1), but the achievable minima
3 (bottom left points) vary significantly by cooling system archetype. The results also indicate a tradeoff
4 between facility-level electricity and water use per workload when selecting data center cooling
5 technologies. That is, data centers with air-cooled chillers and direct expansion systems typically use less
6 water but more electricity per workload, whereas data centers using water-cooled chillers consume less
7 electricity but more water per workload. The implementation of economizers, dry coolers, liquid cooling,
8 or adiabatic cooling in data centers can simultaneously reduce facility-level electricity and water use per
9 workload. Adiabatic cooling, in particular, has been found to significantly reduce electricity use per
10 workload in these settings. However, different combinations of determinants can lead to the same electricity
11 and/or water use per workload. For example, the points marked A have the same facility-level water use
12 per workload, and the points marked B have the same facility-level electricity use per workload),
13 demonstrating again that electricity and water use per workload cannot be simply assessed or compared
14 using a single determinant. In contrast, these results emphasize that there are multiple paths to reducing
15 facility-level electricity and/or water use per workload in data centers.

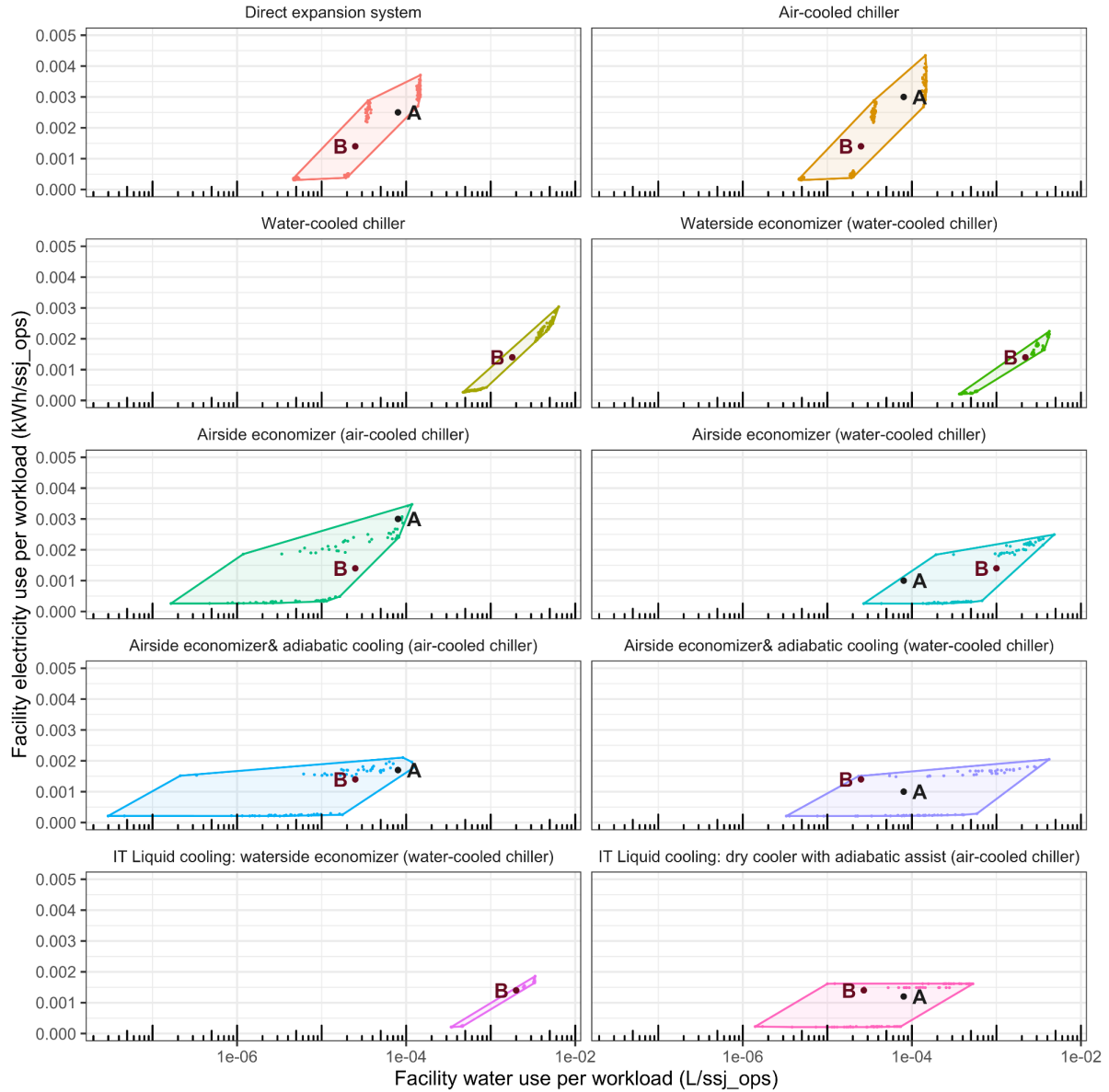


Fig. 5. Relationship between facility electricity and water use per workload by data center cooling system type.

Note: Points A have the same facility water use per workload; all points B have the same facility electricity use per workload.

1

2 3.5. Determinants of WUE-source

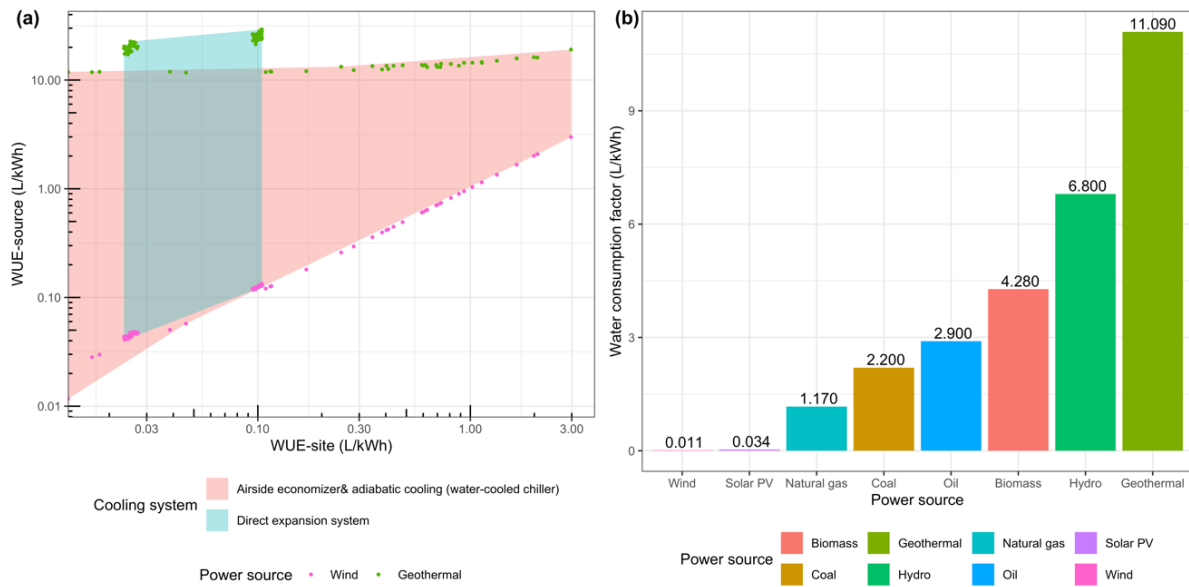
3 WUE-source (a component of Eq. (4)) is often used as a proxy to quantify the overall water use efficiency
 4 (i.e., total water consumption in liters per kWh of electricity consumed) of a data center (Siddik et al.,
 5 2021; The Green Grid, 2011a). To clearly illustrate the determinants of WUE-source, we first compared
 6 the relationship between WUE-source and WUE-site for two common cooling systems (i.e., airside

1 economizer and adiabatic cooling (water-cooled chiller) and direct expansion systems) and two power
2 sources (i.e., wind and geothermal), which results in two regions in Fig. 6 (a). These regions are bounded
3 from left to right by the range of WUE-site values, where the region width corresponding to the airside
4 economizer and adiabatic cooling (water-cooled chiller) is larger than that corresponding to the direct
5 expansion system due to greater variability in WUE-site values (Fig. 4). For a given a cooling technology
6 and a given WUE-site value, the distance between the lower and upper bound of the region is determined
7 by the product of PUE value and WCF (see Eq. (4)), where the lower bound and upper bound is associated
8 with the power source of the lowest U.S. WCF (wind) and the highest U.S. WCF (geothermal), respectively.

9 The WUE-source values of data centers are linearly correlated with both WUE-site values and the WCFs
10 of power sources (Fig. 6 (b)). Together, they define the range of potential WUE-source values for data
11 centers (SI Fig. S8.1). Importantly, large gaps between the lower and upper boundaries of these regions
12 (Fig. 6(a) and SI Fig. S8.1) highlights the huge influence of WCF in determining WUE-source. That is,
13 data centers with high WUE-site values can achieve low WUE-source values by utilizing wind or solar
14 power, while data centers with low WUE-site values can have high WUE-source values due to the use of
15 fossil fuel, hydro, or geothermal based electricity.

16 Fig. 6 (c) depicts the relationship between WUE-site and WUE-source for different data center cooling
17 technologies, with lower and upper vertical bounds associated with minimum and maximal achievable
18 WCFs (Fig. 6(b)), respectively. Due to differences in PUE and WUE-site values for different cooling
19 technologies, the area and shape of the regions vary widely. For example, in Fig. 6 (c), points A and B have
20 the same WUE-site but WUE-source values differ by a factor of 34, while points C and D have the same
21 WUE-source but WUE-site values differ by a factor of 4.5. Depending on the product of the PUE and WCF
22 values, data centers with the same WUE-site value can have very different WUE-source values. The same
23 WUE-source value can be achieved by combining various determinants, which highlights two main
24 approaches for data center WUE-source minimization to address local water issues and water conservation
25 concerns. First, WUE-source can be reduced by improving data center operational efficiency, which can

- 1 reduce both PUE and WUE-site, but the level of this reduction is limited by cooling technology
- 2 implementation and data center climate conditions. Second, WUE-source can be reduced through choice of
- 3 power sources with low WCFs, which is a more general approach that is not constrained by the cooling
- 4 technology implementation, but rather the availability of electric grid options.



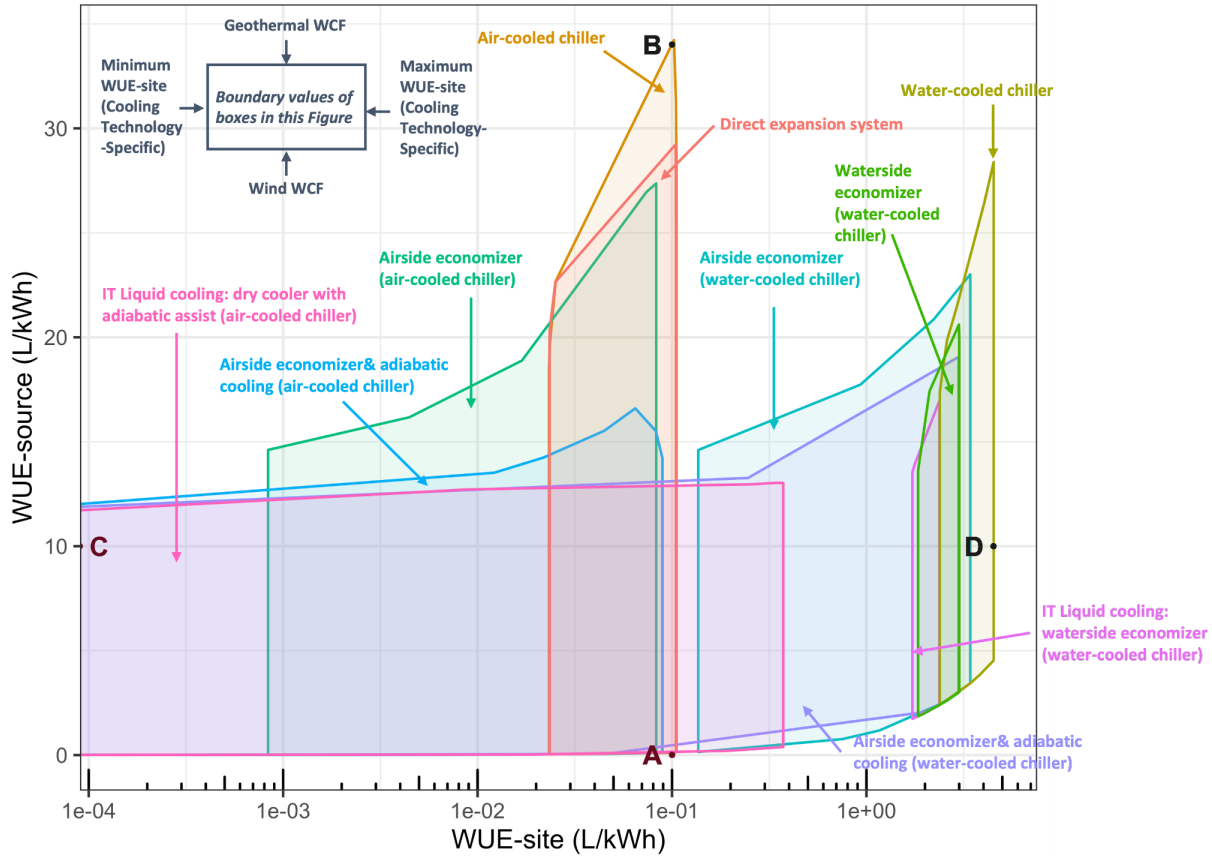


Fig. 6: (a) Relationship between WUE-source and WUE-site in data centers with airside economizer and adiabatic cooling (water-cooled chiller) or direct expansion systems (see SI Fig. S8.1, where the x-axis is linearly spaced); (b) U.S. water consumption factor by power source; (c) Relationship between WUE-source and WUE-site by cooling technology.

3.6. Determinants of overall water use per workload

The final step in our decomposition analysis was to assess, via Eq. (4), the interrelationships discussed in previous sections to quantify overall water use per workload. For simplicity, we analyzed data centers that rely on wind or geothermal power, corresponding to the power sources with the lowest and highest U.S. WCF in Fig. 6 (b) respectively. Fig. 7 (a) shows the relationship between overall and facility-level water use per workload for data centers using airside economizers and using air-cooled chillers as supplemental cooling equipment. Notably, the upper and lower bounds of the regions in Fig. 7 (a) respectively correspond to low and high server workload efficiencies, and the relative distance between them represent the relative importance of server workload efficiency to the overall water use per workload in data centers. Furthermore, the relative distance of the lower and upper bound in Fig. 7 (a) is larger for data centers that use geothermal

power than those use wind power, indicating the significance of reducing overall water use per workload by improving server workload efficiency when a data center is constrained by power sources with high WCFs. Conversely, the relative distance between the lower and upper bound in Fig. 7 (a) is small for data centers that use wind power, highlighting the importance of reducing overall water use by reducing facility-level water use per workload in data centers that rely on low WCF power sources.

Fig. 7 (b) further shows the relationship between the facility-level and the overall water use per workload for data centers with different cooling systems. Data centers with low WCF power typically have a much lower overall water use than those using high WCF power, unless the data center has a significant on-site water consumption (e.g., the data center relies on water-cooled chillers or waterside economizers for cooling), which reconfirms the effectiveness of using low WCF electricity to reduce the water use of data centers. Additionally, the facility-level and overall water use per workload become more positively correlated as data centers transition from high to low WCF power, suggesting that more action should be taken to reduce facility-level water use (either through cooling system selection, PUE or WUE-site reduction, or server workload efficiency optimization) when low-WCF power is already available. Nevertheless, it is again clear that different combinations of determinants (WCF, cooling system type, and efficiency, etc.) can lead to similar overall water use per workload values, so data centers should not be compared on the basis of WUE-source, WCF, or WUE-site alone. This finding is again illustrated via the region between the two dashed lines in Fig. 7 (b).

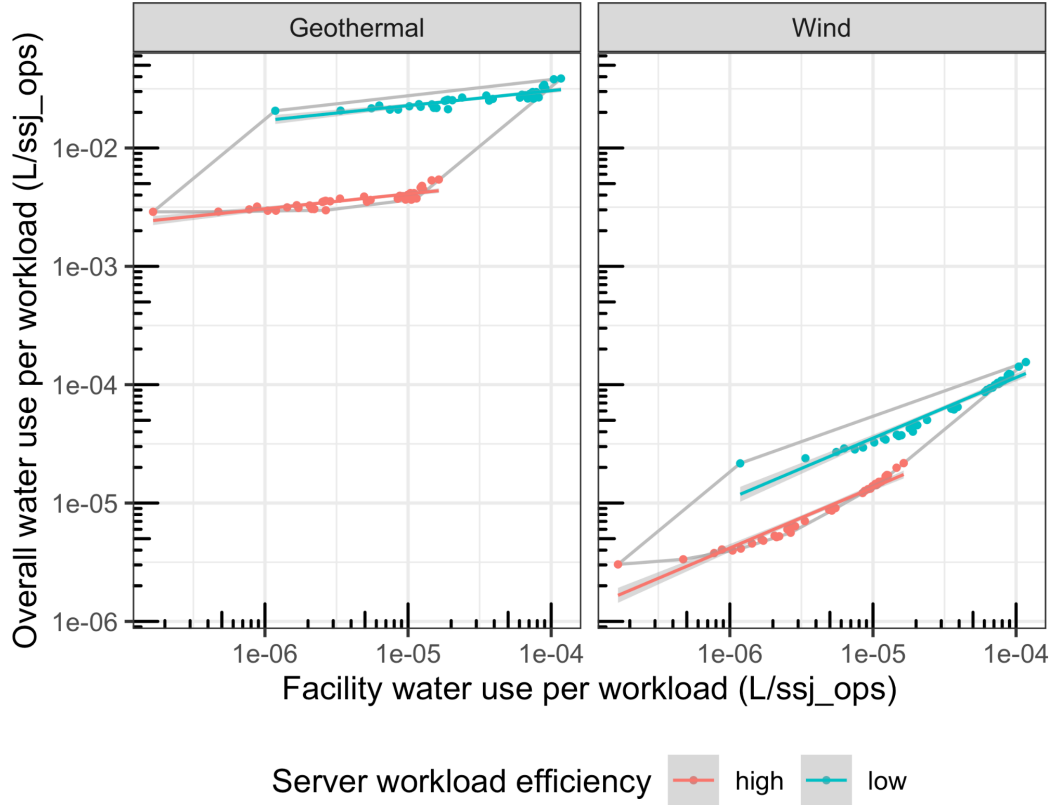


Fig. 7 (a): Relationship between data center facility-level and overall water use per workload for data centers with airside economizers (air-cooled chillers) and rely on geothermal or wind power.
 Note: The regions were drawn by inputting the minimum and maximum server electricity per workload values from Fig. 2, along with the PUE and WUE-site values from Fig. 4 and the WCFs for geothermal and wind from Fig. 6 (b), into Eq. (4).

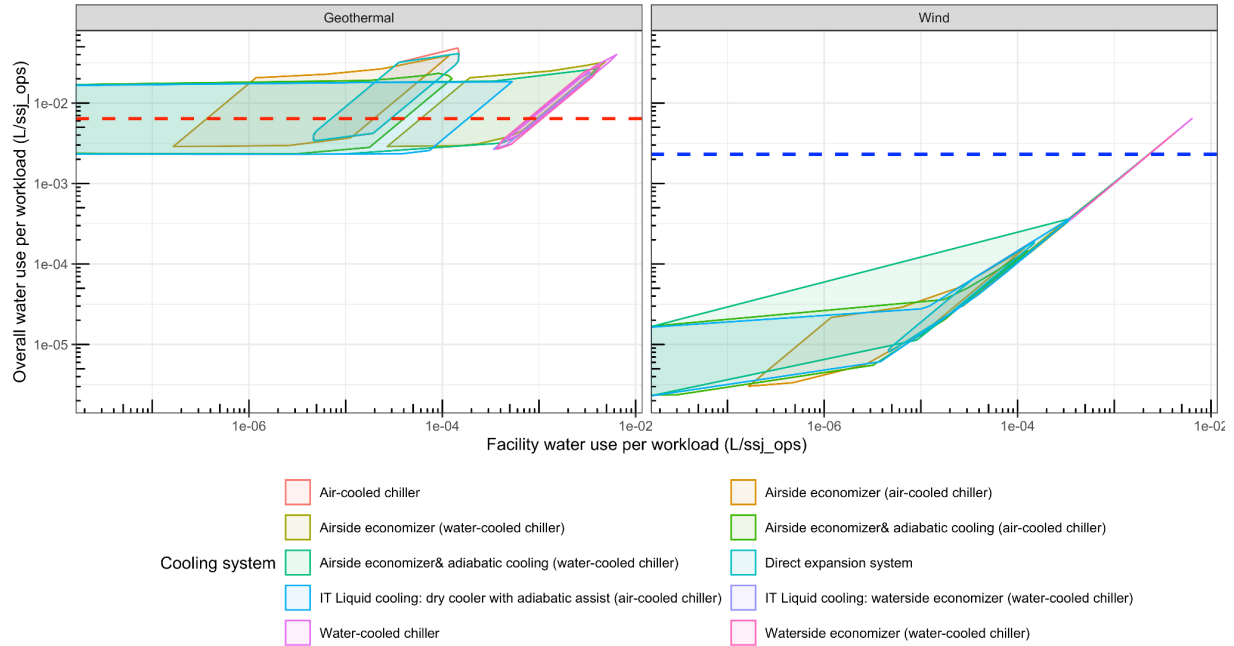


Fig. 7 (b): Relationship between data center facility-level and overall water use per workload by cooling technology for data centers that rely on geothermal/wind power (see SI Fig. S9.1 for data centers that rely on other power sources). Blue dashed line ~ minimum overall water use per workload for data centers using geothermal power; Red dashed line ~ maximum overall water use per workload for data centers using wind power.

3.7. Sensitivity analysis

Fig. 8. shows the sensitivity results based on Sobol's total effect index (Sobol, 2001), which ranks the relative importance of the determinants of overall water use per workload via variance decomposition. Server efficiency quantile and server utilization level are the first and third largest contributors to variance in the overall water use result, as they have large impact on direct server electricity use per workload, which signifies that overall data center water use per workload can also be reduced through installing power-efficient servers (e.g., Energy Star certified servers) or server workload management (e.g., consolidation and virtualization). The WCF is the second largest contributor to the variance in the overall water use per workload results, suggesting that powering the data center with low-WCF electricity such as wind or solar PV can be effective in reducing overall data center water use. Cooling system type, data center infrastructure operating efficiency, and climatic conditions are important factors in determining data center PUE and WUE-site, making them non-trivial factors in data center overall water use minimization. Inactive server percentage and server fresh cycle also affect the overall water use per workload through affecting server workload efficiency, further confirming the usefulness of server management through removal of needless equipment or equipment refurbishment.

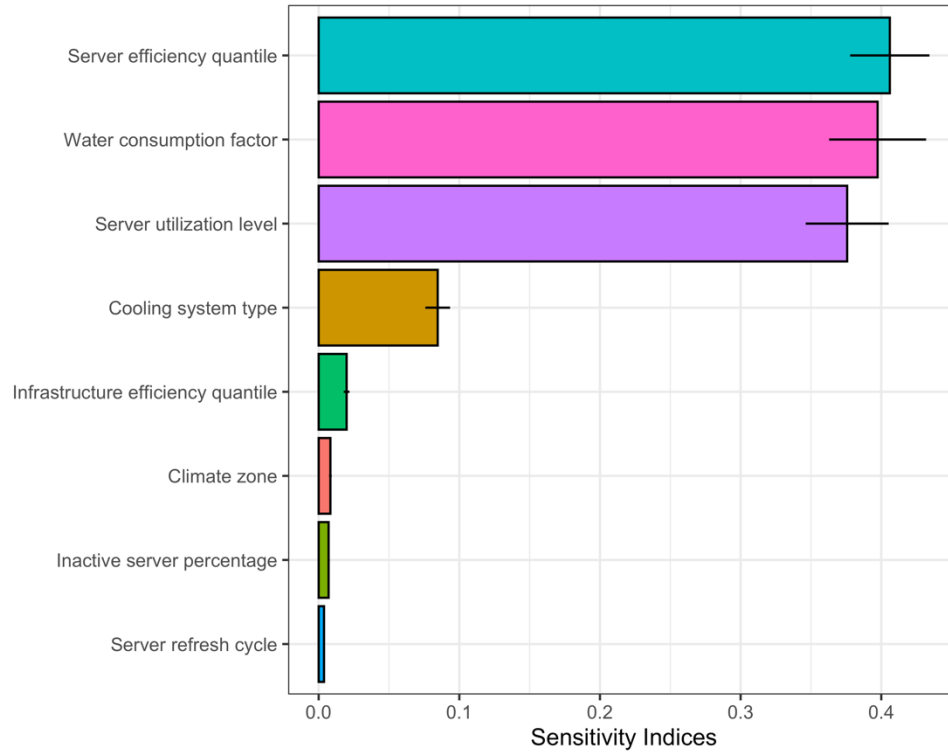


Fig. 8. Importance ranking (based on Sobol's total-effect index) for all of the determinants of the overall water use per workload (Error bar ~ 95% confidence interval of sensitivity indices).

Note: Server efficiency quantile ~ measures the server power efficiency for a given year (SI S2); infrastructure efficiency quantile ~ measures the operating efficiency of the data center infrastructure (i.e., PUE and WUE-site) for a given cooling technology under a given climate zone.

4. Conclusion

Increasing digitalization and freshwater restrictions have brought data center water conservation into the spotlight. While more data center operators are beginning to track and report WUE-site and WUE-source, these metrics fall short in measuring water use at the workload level, which is the ideal basis for understanding and comparing water use in data centers.

This study focuses on analyzing the determinants of overall workload-level water use, enabling more direct comparisons across data centers with diverse site-specific characteristics. Using statistical analysis of recent server performance data, we first highlighted the relationship between server electricity efficiency, utilization levels, refresh rates, and inactive server management as determinants of direct server electricity use per workload. Next, we examined the relationship between PUE and WUE-site through climate- and

1 technology-specific simulations of major data center cooling archetypes, analyzing their influence on
2 facility-level electricity and water use per workload. Finally, we assessed the relationship between electrical
3 WCFs, WUE-source, and overall water use per workload. In general, this paper addresses critical gaps in
4 existing research and contributes in the following key ways:

- 5 • Our study provides a comprehensive workload-level water use metric, integrating traditional
6 efficiency measures such as PUE, WUE-site, WCF, and server workload efficiency. This metric
7 enables consistent and comparable evaluations of data centers with varying site-specific
8 characteristics, offering a path toward standardizing industry reporting, which is currently
9 inconsistent in scope, quality, and granularity.
- 10 • Unlike prior studies that focus primarily on either direct or indirect water use, this paper provides
11 a holistic analysis, covering direct water use such as space humidification, adiabatic cooling, and
12 cooling tower operations, as well as indirect water use, including upstream water required for
13 electricity fuel production, processing, and transportation, and water use at the point of electricity
14 generation. This comprehensive approach lays the groundwork for developing integrated water-
15 saving strategies.
- 16 • We conduct an extensive parametric analysis of workload-level water use, considering various
17 determinants such as server efficiency, electrical grid water consumption factors, server utilization,
18 cooling system type, infrastructure efficiency, climate zone, inactive server percentages, and server
19 refresh cycles. This analysis provides actionable insights and tailored strategies for reducing water
20 use under diverse data center conditions.
- 21 • Uniquely, this paper incorporates server workload efficiency and its determinants into the analysis
22 of data center water use per workload. This provides a foundational framework to understand and
23 improve water use for different digital services, supporting policymakers and operators as they
24 increasingly focus on the direct and indirect water consumption of digital services.

5. Implications and limitations

Our findings can help policymakers design common measures to promote water conservation in data centers. For example, further promoting renewable energy (wind or solar PV), the use of the most efficient servers, and techniques such as virtualization for maintaining high levels of server utilization can effectively reduce water uses at the national level, while maximizing service outputs. Our findings also highlight that there is often no “silver bullet” for addressing the overall water use associated with all types of data centers, so strategies should be developed on a case-by-case basis. For instance, if the data center is geographically constrained with respect to its electric grid or climate, increasing average server workload efficiency is the most effective strategy to reduce water use. Furthermore, judicious selection of cooling technologies during the design phase of a data center has great potential to reduce a data center’s water use, which can lessen competition for local water resources from data center on-site water consumption in the meantime.

Our analysis also illustrates the value of evaluating water use at the workload hosting level in data centers, which brings to light the necessity of considering the computing power efficiency of IT equipment. This is an aspect that is frequently overlooked by sustainability analysts and policymakers, who have historically focused on PUE and WUE metrics. Our analysis enables impartial comparisons of electricity and water consumption among data centers by proposing several workload-based indicators, including direct server electricity use per workload, facility-level water use per workload, and overall water use per workload. For example, a cloud data center that utilizes airside economizers and adiabatic cooling, in addition to using water-cooled chillers for supplementary cooling, may consume more water on-site per kilowatt-hour of electricity used compared to a traditional data center with a direct-expansion system. However, such a comparison may result in a biased view that the traditional data center is more water-efficient by ignoring the number of digital services that can be delivered for the same amount of on-site and off-site water use. Our analysis fills the gap in the current understanding of water uses in data centers by providing examples of different scenarios that result in the same overall water use per workload.

1 While the primary focus of this study is the overall water use of data centers, the conservation strategies we
2 suggest also have the potential to impact other aspects of data center operations. Data center workload
3 management through virtualization allows multiple applications to run on a single server, potentially
4 reducing the required IT capacity and saving initial investment costs through proper capacity planning.
5 Improving the efficiency of IT equipment and infrastructure can reduce the electricity and water usage per
6 data service, leading to cost savings (i.e., electric and water bills) in data center operations. Increasing
7 energy efficiency and using renewable energy sources like wind or solar power can also reduce carbon
8 emissions and contribute to climate change mitigation efforts, which are crucial in the current global
9 warming context. To summarize, reducing the water use of data centers has many co-benefits such as
10 reduced resource usage, reduced electric and water costs, and reduced carbon emissions. This provides
11 strong evidence for policy incentives and operational measures to further improve water use per workload
12 in data centers.

13 There are several limitations to consider in this analysis. Because of limited information on server
14 computational efficiency, we rely on the SPECpower_ssj 2008 benchmark database, which may not fully
15 reflect the distribution of server computational efficiency in all data center operations over time. There are
16 concerns that even the least efficient servers in the SPECpower_ssj 2008 benchmark database may be more
17 efficient than typical volume servers on the market. This could result in an underestimation of the upper
18 bounds of water uses in our analysis. Our study primarily examines servers as the main component of IT
19 loads, as they are the primary hosts for workloads. However, future research could consider a broader range
20 of IT loads, including graphics processing units (GPUs), storage, and network loads to provide a more
21 comprehensive assessment. Additionally, the WCFs analyzed in this study are based on generation-
22 weighted U.S. averages for different power sources, without accounting for variations by region, cooling
23 technology, prime mover technology, and other factors (Lee et al., 2018). It is also important to further
24 emphasize that the workload-level water use analyzed here takes into account only water consumption, not
25 water withdrawals, which are also important for understanding the potential effect that data centers can

1 have on local water resources. We recommend that decision makers consider using electric power water
2 withdrawal factors in combination with the WCFs we used to assess total site water use in the future.
3 Finally, this paper focuses on dissecting and analyzing the determinants of workload-level water use in data
4 centers, a metric sometimes referred to as the water footprint in data center sustainability research (Lei et
5 al., 2023; Ristic et al., 2015; Siddik et al., 2021). For example, (Aldaya et al., 2012; Ansorge and
6 Stejskalová, 2025) have underscored the importance of adhering to precise definitions of the water footprint
7 to avoid confusion in critical water studies. As interest in data center water use and footprint research grows,
8 establishing a clear and standardized definition of the data center water footprint, specifically tailored to
9 this industry, is essential to guide future research and policy development.

11 Acknowledgements

12 N.L. and E.M. express gratitude for support from Leslie and Mac McQuown, which funded early phases of
13 this research conducted at Northwestern University, and for support from the Mellichamp Academic
14 Initiative in Sustainability at University of California, Santa Barbara. N.L. and A.S. acknowledge that
15 portions of this work was conducted at Lawrence Berkeley National Laboratory and supported by the U.S.
16 Department of Energy (DOE) Industrial Efficiency and Decarbonization Office (IEDO) under Contract No.
17 DE-AC02-05CH11231. The views and opinions of the authors expressed herein do not necessarily state or
18 reflect those of the United States Government or any agency thereof, or The Regents of the University of
19 California.

20 Supporting Information

21 Supporting information to this article can be found online. The supporting data and code associated with this
22 article can be found at:

23 [https://github.com/nuoaleon/The-Water-Use-of-Data-Center-Workloads-A-Review-and-Assessment-of-](https://github.com/nuoaleon/The-Water-Use-of-Data-Center-Workloads-A-Review-and-Assessment-of-Key-Determinants)
24 [Key-Determinants](https://github.com/nuoaleon/The-Water-Use-of-Data-Center-Workloads-A-Review-and-Assessment-of-Key-Determinants)

References

- Aldaya, M.M., Chapagain, A.K., Hoekstra, A.Y., Mekonnen, M.M., 2012. The Water Footprint Assessment Manual, 0 ed. Routledge. <https://doi.org/10.4324/9781849775526>
- Andrae, A.S.G., Edler, T., 2015. On Global Electricity Usage of Communication Technology: Trends to 2030. *Chall.* 2015 Vol 6 Pages 117-157 6, 117–157. <https://doi.org/10.3390/CHALLE6010117>
- Ansorge, L., Stejskalová, L., 2025. Water footprint which is not the water footprint: Critical review of the article by Müller et al. (2024). *J. Environ. Manage.* 374, 124038. <https://doi.org/10.1016/j.jenvman.2025.124038>
- ASHRAE, 2014. Liquid Cooling Guidelines for Datacom Equipment Centers, 2nd Edition, ASHRAE Datacom Series. Atlanta.
- ASHRAE, 2013. Climatic data for building design standards. ASHRAE Stand.
- Barroso, L.A., Hölzle, U., 2007. The case for energy-proportional computing. *Computer* 40, 33–37. <https://doi.org/10.1109/MC.2007.443>
- Belkhir, L., Elmeligi, A., 2018. Assessing ICT global emissions footprint: Trends to 2040 & recommendations. *J. Clean. Prod.* 177, 448–463. <https://doi.org/10.1016/J.JCLEPRO.2017.12.239>
- Evans, T., 2004. The different types of air conditioning equipment for IT environments. APC White Pap. 24.
- Facebook, 2014. Open sourcing PUE/WUE dashboards [WWW Document]. Facebook Eng. URL <https://engineering.fb.com/2014/03/14/data-center-engineering/open-sourcing-pue-wue-dashboards/> (accessed 7.13.21).
- Google, 2024. Environmental Report - Google Sustainability [WWW Document]. Sustainability. URL <https://sustainability.google/reports/google-2024-environmental-report/> (accessed 1.20.25).
- Gozcu, O., Ozada, B., Carfi, M.U., Erden, H.S., 2017. Worldwide energy analysis of major free cooling methods for data centers, in: Proceedings of the 16th InterSociety Conference on Thermal and Thermomechanical Phenomena in Electronic Systems, ITHERM 2017. <https://doi.org/10.1109/ITHERM.2017.7992592>
- Grubert, E., Sanders, K.T., 2018. Water Use in the United States Energy System: A National Assessment and Unit Process Inventory of Water Consumption and Withdrawals. *Environ. Sci. Technol.* <https://doi.org/10.1021/acs.est.8b00139>
- Horner, N., Azevedo, I., 2016. Power usage effectiveness in data centers: overloaded and underachieving. *Electr. J.* 29, 61–69. <https://doi.org/10.1016/J.TEJ.2016.04.011>
- Iyengar, M., David, M., Parida, P., Kamath, V., Kochuparambil, B., Graybill, D., Schultz, M., Gaynes, M., Simons, R., Schmidt, R., Chainer, T., 2012. Server liquid cooling with chiller-less data center design to enable significant energy savings. *Annu. IEEE Semicond. Therm. Meas. Manag. Symp.* 212–223. <https://doi.org/10.1109/STHERM.2012.6188851>
- Jones, N., 2018. How to stop data centres from gobbling up the world’s electricity. *Nature* 561, 163–166. <https://doi.org/10.1038/D41586-018-06610-Y>
- Lee, U., Han, J., Elgowainy, A., Wang, M., 2018. Regional water consumption for hydro and thermal electricity generation in the United States. *Appl. Energy* 210, 661–672. <https://doi.org/10.1016/j.apenergy.2017.05.025>
- Lei, N., 2022. A Hybrid Physics-Based and Data-Driven Modeling Framework for Energy and Water Use Analysis of Data Centers with Spatio-Temporal Resolution. Northwestern University, Evanston, IL, USA.
- Lei, N., Lu, J., Cheng, Z., Cao, Z., Shehabi, A., Masanet, E., 2023. Geospatial assessment of water footprints for hyperscale data centers in the United States. *J. Phys. Conf. Ser.* 2600, 172003. <https://doi.org/10.1088/1742-6596/2600/17/172003>
- Lei, N., Masanet, E., 2022. Climate- and technology-specific PUE and WUE estimations for U.S. data centers using a hybrid statistical and thermodynamics-based approach. *Resour. Conserv. Recycl.* 182, 106323. <https://doi.org/10.1016/J.RESCONREC.2022.106323>
- Lei, N., Masanet, E., 2021. Global Data Center Energy Demand and Strategies to Conserve Energy, in: Data Center Handbook, 2nd Edition. WILEY, Palo Alto, CA. <https://doi.org/10.1002/9781119597537.ch2>

1 Lei, N., Masanet, E., 2020. Statistical analysis for predicting location-specific data center PUE and its
 2 improvement potential. *Energy* 201, 117556. <https://doi.org/10.1016/j.energy.2020.117556>
 3 Masanet, E., Lei, N., Koomey, J., 2024. To better understand AI's growing energy use, analysts need a data
 4 revolution. *Joule* 8, 2427–2436. <https://doi.org/10.1016/j.joule.2024.07.018>
 5 Masanet, E., Shehabi, A., Koomey, J., 2013. Characteristics of low-carbon data centres. *Nat. Clim. Change* 3,
 6 627–630. <https://doi.org/10.1038/nclimate1786>
 7 Masanet, E., Shehabi, A., Lei, N., Smith, S., Koomey, J., 2020. Recalibrating global data center energy-use
 8 estimates. *Science* 367, 984–986. <https://doi.org/10.1126/science.aba3758>
 9 Meta Sustainability, 2023. Sustainability Report [WWW Document]. URL [https://sustainability.fb.com/2023-](https://sustainability.fb.com/2023-sustainability-report/)
 10 [sustainability-report/](https://sustainability.fb.com/2023-sustainability-report/) (accessed 4.22.24).
 11 Microsoft, 2024. Environmental Sustainability Report Data Fact Sheet [WWW Document].
 12 MLCommons, 2024. Benchmark MLPerf Inference: Datacenter [WWW Document]. URL
 13 <https://mlcommons.org/benchmarks/inference-datacenter/> (accessed 4.22.24).
 14 Mytton, D., 2021. Data centre water consumption. *Npj Clean Water* 2021 41 4, 1–6.
 15 <https://doi.org/10.1038/s41545-021-00101-w>
 16 Oregon Tech, 2022. Google's water use is soaring in The Dalles, records show, with two more data centers to
 17 come [WWW Document]. *oregonlive*. URL [https://www.oregonlive.com/silicon-](https://www.oregonlive.com/silicon-forest/2022/12/googles-water-use-is-soaring-in-the-dalles-records-show-with-two-more-data-centers-to-come.html)
 18 [forest/2022/12/googles-water-use-is-soaring-in-the-dalles-records-show-with-two-more-data-](https://www.oregonlive.com/silicon-forest/2022/12/googles-water-use-is-soaring-in-the-dalles-records-show-with-two-more-data-centers-to-come.html)
 19 [centers-to-come.html](https://www.oregonlive.com/silicon-forest/2022/12/googles-water-use-is-soaring-in-the-dalles-records-show-with-two-more-data-centers-to-come.html) (accessed 12.27.22).
 20 Peer, R.A.M., Grubert, E., Sanders, K.T., 2019. A regional assessment of the water embedded in the US
 21 electricity system. *Environ. Res. Lett.* <https://doi.org/10.1088/1748-9326/ab2daa>
 22 Ristic, B., Madani, K., Makuch, Z., 2015. The water footprint of data centers. *Sustain. Switz.* 7.
 23 <https://doi.org/10.3390/su70811260>
 24 Ryckbosch, F., Polfliet, S., Eeckhout, L., 2011. Trends in server energy proportionality. *Computer* 44, 69–72.
 25 <https://doi.org/10.1109/MC.2011.130>
 26 Scaleway, 2021. Environmental leadership [WWW Document]. URL
 27 <https://www.scaleway.com/en/environmental-leadership/> (accessed 7.17.21).
 28 Sharma, R., Shah, A., Bash, C., Christian, T., Patel, C., 2009. Water efficiency management in datacenters:
 29 Metrics and methodology, in: 2009 IEEE International Symposium on Sustainable Systems and
 30 Technology, ISSST '09 in Cooperation with 2009 IEEE International Symposium on Technology and
 31 Society, ISTAS. <https://doi.org/10.1109/ISSST.2009.5156773>
 32 Sharma, R., Shah, A.J., Bash, C., Christian, T., Patel, C., 2008. Water Efficiency Management in Datacenters
 33 (Part I): Introducing a water usage metric based on available energy consumption. *TR HP Labs*.
 34 Shehabi, A., Smith, S.J., Hubbard, A., Newkirk, A., Nuoa, L., Siddik, M.A.B., Jonathan, K., Eric, M., Dale, S.,
 35 2024. 2024 United States Data Center Energy Usage Report. Berkeley Lab.
 36 Shehabi, A., Smith, S.J., Sartor, D.A., Brown, R.E., Herrlin, M., Koomey, J.G., Masanet, E.R., Horner, N.,
 37 Azevedo, I.L., Lintner, W., 2016. United States Data Center Energy Usage Report. Berkeley Lab 65.
 38 Siddik, M.A.B., Shehabi, A., Marston, L.T., 2021. The environmental footprint of data centers in the United
 39 States. *Environ. Res. Lett.* <https://doi.org/10.1088/1748-9326/abfba1>
 40 Siddik, M.A.B., Shehabi, A., Rao, P., Marston, L.T., 2024. Spatially and Temporally Detailed Water and Carbon
 41 Footprints of U.S. Electricity Generation and Use. *Water Resour. Res.* 60, e2024WR038350.
 42 <https://doi.org/10.1029/2024WR038350>
 43 Sobol, I.M., 2001. Global sensitivity indices for nonlinear mathematical models and their Monte Carlo
 44 estimates. *Math. Comput. Simul.* 55, 271–280. [https://doi.org/10.1016/S0378-4754\(00\)00270-6](https://doi.org/10.1016/S0378-4754(00)00270-6)
 45 Standard Performance Evaluation Corporation, 2022. SPECpower_ssj® 2008 [WWW Document]. URL
 46 https://www.spec.org/power_ssj2008/ (accessed 6.22.22).
 47 The Green Grid, 2011a. Water usage effectiveness (WUE): a Green Grid data center sustainability metric.
 48 White Pap. 35.
 49 The Green Grid, 2011b. PUE: The Green Grid metric for evaluating the energy efficiency in DC (Data Center).
 50 Measurement method using the power demand. *INTELEC Int. Telecommun. Energy Conf. Proc.*

The Water Use of Data Center Workloads: A Review and Assessment of Key Determinants

Supplementary Information (SI)

Nuoa Lei ¹, Jun Lu ², Arman Shehabi ¹, Eric Masanet ^{3,4}

¹ Energy Technologies Area, Lawrence Berkeley National Laboratory, Berkeley, CA, USA

² Division of Epidemiology and Biostatistics, University of Illinois, Chicago, IL, USA

³ Bren School of Environmental Science and Management, University of California, Santa Barbara, CA, USA

⁴ Department of Mechanical Engineering, University of California, Santa Barbara

S1: Server Vintage Distributions

A comprehensive factorial analysis was conducted to investigate how the refresh cycle affects water use per workload in data centers ($\frac{liters}{ssj_ops}$). The study encompassed refresh cycles spanning from 1 to 8 years, representing a spectrum of practices from small-scale traditional data centers to hyperscale facilities (Horizon Editorial, 2023; Supermicro, 2019). Different refresh cycle durations can be mapped to different distributions of server vintage, as outlined in Table S1 below.

Table S.1. Relationship between server vintage distribution and refresh cycle (reference year: 2023).

Server Vintage Distributions v.s. Refresh Cycle								
Refresh Cycle	Server Vintage Distributions							
Length (Years)	2016	2017	2018	2019	2020	2021	2022	2023
1	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	100.0%
2	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	50.0%	50.0%
3	0.0%	0.0%	0.0%	0.0%	0.0%	33.3%	33.3%	33.3%
4	0.0%	0.0%	0.0%	0.0%	25.0%	25.0%	25.0%	25.0%
5	0.0%	0.0%	0.0%	20.0%	20.0%	20.0%	20.0%	20.0%
6	0.0%	0.0%	16.7%	16.7%	16.7%	16.7%	16.7%	16.7%
7	0.0%	14.3%	14.3%	14.3%	14.3%	14.3%	14.3%	14.3%
8	12.5%	12.5%	12.5%	12.5%	12.5%	12.5%	12.5%	12.5%

1 **S2: Server Power Use And Power-performance**

2 In this study, we utilized the "SPECpower_ssj 2008" database (Standard Performance Evaluation
3 Corporation, 2022) to estimate server power usage and performance. The "SPECpower_ssj 2008"
4 database is updated quarterly and comprises submissions from various server vendors and
5 organizations. It includes data on power usage and performance metrics of volume servers tested
6 using the "SPECpower_ssj 2008" benchmark, which is the industry's first standardized benchmark
7 designed to assess the power efficiency characteristics of volume servers. Since 2007, the
8 "SPECpower_ssj 2008" database has provided consistent and standardized records of server power
9 usage and performance, offering insights into electricity consumption per server workload in data
10 centers over the years.

11 Based on the "SPECpower_ssj 2008" database, the local linear kernel regression (LLKR) model
12 (Kafadar et al., 1999) was employed to construct response surfaces for both server power and
13 server power-performance. This forms the foundation for analyzing average server power and
14 power-performance, considering factors such as server vintage distribution, average server
15 utilization level, and fraction of inactive servers (Eqs. (5) ~ (8)). LLKR, a type of non-parametric
16 regression model, is chosen for its flexibility in handling unexpected nonlinear relationships
17 between inputs and outputs. Near a given location x , LLKR adopts the form of Eq. (S1) for any
18 predictor X_k .

$$19 \quad Y_k = E(X_k) + \epsilon_k = m_L(x) + m'_L(x)(X_k - x) + \epsilon_k \quad (S1)$$

20 where Y_k represents the response variable, ϵ_k denotes the random error term satisfying $E(X_k) = 0$,
21 and $m_L(x) \approx m(x) + \frac{1}{2}m''(x)h^2$ is termed the local linear kernel estimator. In this expression,
22 $m(x)$, defined by Eq. (S2) below, represents a differentiable function commonly known as the
23 local constant kernel estimator.

$$m(x) = \frac{\sum_{k=1}^n K\left(\frac{x-X_k}{h}\right)Y_k}{\sum_{k=1}^n K\left(\frac{x-X_k}{h}\right)}, -\infty < x < +\infty \quad (S2)$$

where $K(\cdot)$ represents a kernel function, with the Gaussian kernel function chosen for this study (as per Eq. (S3)). The parameter h governs the extent of smoothing and was estimated using the Akaike Information Criterion (Hurvich et al., 1998).

$$K(d) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}\|d\|^2}, -\infty < d < +\infty \quad (S3)$$

We developed LLKR models to predict server power (P) and power-performance ($Perf$) using a predictor vector comprising server utilization level (u) and server installation year (y). To examine the impact of server power efficiency on data center water use per workload, we constructed response surfaces for both P and $Perf$ based on servers ranked at the 5th and 95th quantiles of overall power efficiency (Standard Performance Evaluation Corporation, 2022) respectively each year in the “SPECpower_ssj 2008” database. Using the `sm.regression` function in R’s ‘sm’ package (Bowman, 2018), we built the associated LLKR models, corresponding to $P(\cdot)$ and $Perf(\cdot)$ in Eqs. (6) – (8).

Fig. S.2.1 (A) and (B) display the resulting response surfaces for server power and server power-performance, revealing several observations: (1) server power generally rises with utilization levels; (2) more efficient servers tend to exhibit lower idle power consumption; (3) server power-performance is $0 \frac{ssj_ops}{W}$ at idle state, increasing with both utilization level and server installation year, yet the gap in power-performance between the least and most efficient servers widens over time; and (4) the relationships between responses and predictors display nonlinear patterns, underscoring the suitability of our nonparametric regression model for this study.

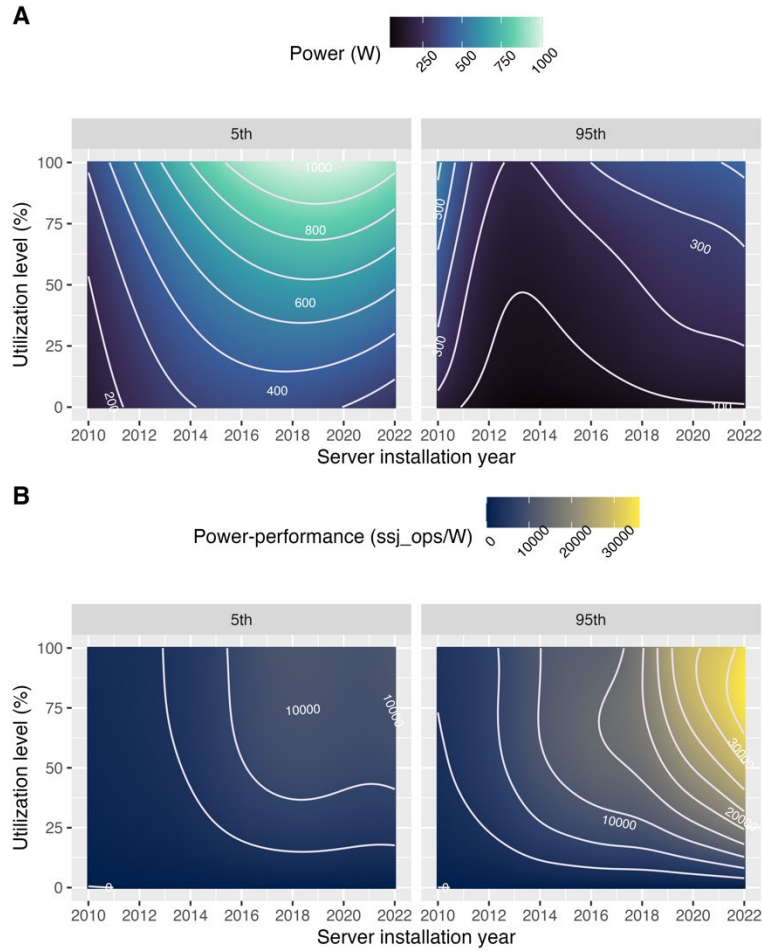


Fig. S2.1 Response surfaces for (A) server power versus server utilization level and server installation year; and (B) server power-performance versus server utilization level and server installation year (5th efficient ~ low server power efficiency at 5th quantile; 95th efficient ~ high server power efficiency at 95th quantile).

S3: PUE And WUE-site

To evaluate the interrelationships between PUE and WUE-site values and elucidate how their determinants influence data center water use per workload, we conducted a thorough review of the simulation methodology and results for PUE and WUE-site (Lei et al., 2023; Lei and Masanet,

2022, 2020), complemented by additional simulations of cooling systems. In total, we examined 10 different systems, encompassing the most typical data center cooling technologies (Evans, 2004) for both air-cooled and liquid-cooled information technology (IT) equipment, as outlined in Table S3.1 below.

Table S3.1. Summary of cooling systems considered in this work.

Cooling System	System Summary
Direct expansion system	Direct Expansion systems, typically called computer room air conditioners (CRACs) are frequently utilized in data centers due to their simplicity and cost-effectiveness. In such systems, refrigerant flows directly through indoor coils, cooling the air within the data center.
Air-cooled chiller	Air-cooled chillers are widely utilized across data centers of varying sizes. These systems employ air-cooled condensers to remove heat from the system and are commonly paired with Computer Room Air Handler (CRAH) units to maintain ideal IT operating conditions. They are preferred options in situations where water conservation is crucial or where space limitations prevent the installation of cooling towers.
Water-cooled chiller	Water-cooled chillers are widely utilized in data centers owing to their high efficiency and capacity to manage substantial cooling requirements. These systems utilize water-cooled condensers to extract heat from the system, subsequently releasing it into the environment via

	cooling towers. They are typically integrated with CRAH units to ensure optimal IT operating conditions. However, the water evaporation process in cooling towers has raised concerns regarding data center water consumption.
Waterside economizer (water-cooled chiller)	This configuration incorporates a waterside economizer system in data centers to minimize energy consumption and operational costs by utilizing cool water from natural resources (such as lakes, rivers, or the sea) or produced by cooling towers when the outdoor air is cool and dry, thereby reducing reliance on mechanical refrigeration. In instances where the waterside economizer is insufficient, a water-cooled chiller provides supplementary cooling. However, it's worth noting that this system may consume significant amounts of water in its evaporative cooling towers.
Airside economizer (air-cooled chiller)	This setup utilizes an airside economizer system, which leverages outdoor air to cool the internal space of a data center during favorable weather conditions. This approach effectively reduces the data center's cooling energy consumption and operational costs. Additionally, an air-cooled chiller serves as supplementary cooling when the airside economizer alone cannot meet the cooling demands.
Airside economizer (water-cooled chiller)	Similar to the airside economizer (air-cooled chiller), but this setup incorporates a water-cooled chiller system to provide supplementary cooling during heat waves or periods of increased cooling demands

	when the airside economizer alone cannot meet the cooling requirements.
Airside economizer & adiabatic cooling (air-cooled chiller)	Similar to the airside economizer (air-cooled chiller), this configuration seeks to minimize or eliminate mechanical refrigeration by leveraging the adiabatic process, wherein water evaporates directly into the air to provide cooling. This approach is highly energy-efficient and cost-effective, and in certain climate zones, it can entirely replace the need for supplementary cooling throughout the year. Additionally, this setup accounts for the potential use of an air-cooled chiller system when supplementary cooling is necessary.
Airside economizer & adiabatic cooling (water-cooled chiller)	Similar to the airside economizer & adiabatic cooling (air-cooled chiller), but this setup incorporates a water-cooled chiller system to provide supplementary cooling during heat waves or periods of increased cooling demands when the airside economizer and adiabatic cooling together cannot meet the cooling requirements.
Liquid cooling: waterside economizer (water-cooled chiller)	This setup utilizes a waterside economizer system in data centers to decrease dependence on mechanical refrigeration, which is well-suited for liquid cooling (e.g., rear door heat exchanger, direct-to-chip, and immersion). Because liquid-cooled IT systems can often be operated at higher water/refrigerant temperatures than air-cooled IT systems, waterside economizers effectively fulfill data center cooling requirements for much of the year, especially in regions with favorable

	climates. When necessary, a water-cooled chiller system is deployed for supplementary cooling. While highly energy-efficient, this system may consume substantial amounts of water in its evaporative cooling towers.
Liquid cooling: dry cooler with adiabatic assist (air-cooled chiller)	The setup is designed to minimize reliance on mechanical refrigeration by utilizing dry coolers that are supported by the adiabatic cooling option, which is highly compatible with liquid cooling systems including rear door heat exchanger, direct-to-chip, and immersion liquid cooling methods. Given the higher facility water/refrigerant temperatures of liquid-cooled IT systems compared to air-cooled ones, dry coolers can effectively meet data center cooling needs for much of the year, particularly in regions with favorable climates. The dry cooler operations involve two distinct modes. In dry mode, the setup efficiently rejects heat using ambient air, while in wet mode, adiabatic pre-cooling enhances cooling efficiency by spraying or circulating water through pads upstream of the heat exchanger coils. This setup adapts to changing weather conditions dynamically, reducing energy consumption while conserving water compared to using evaporative cooling towers. An air-cooled chiller system is deployed for supplementary cooling when needed.

- 1 **Note:** parentheses denote a data center's supplementary cooling system and may not actually be
- 2 deployed in favorable climate zones. For example, in the scenario of an airside economizer (air-
- 3 cooled chiller), the air-cooled chiller would only be utilized when the airside economizer alone

cannot meet the data center's cooling needs; systems whose names do not begin with 'Liquid cooling' represent air-cooled IT systems.

For each cooling system outlined in Table S3.1 within a specified climate zone, we systematically generate 50 scenarios depicting the interplay between its PUE and WUE-site. These scenarios are derived from uncertain facility system variables and typical meteorological year (TMY) data for the representative city of the respective climate zone, as illustrated in Fig. S3.1. Additionally, Table S3.2 provides a summary of the sources of the pertinent model and data utilized for the scenario generation.

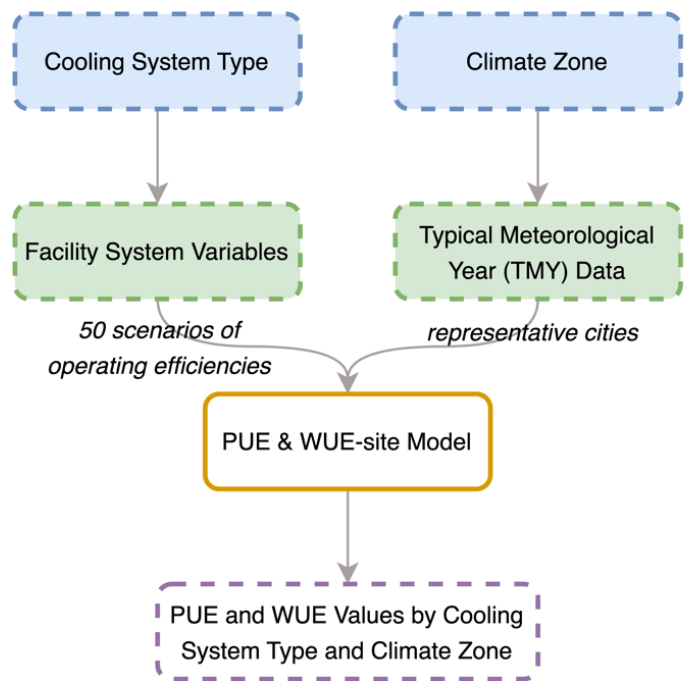


Fig. S3.1. Simplified diagram of PUE and WUE-site scenario generation.

Table S3.2. Summary of data and model source used for PUE and WUE-site scenario generation.

Model and Inputs	Notes	Source
Physical PUE & WUE-site Model	The PUE model accounts for all power-consuming elements within the data center, encompassing but not limited to information technology equipment, power supply and distribution losses, lighting, and notably, the power consumption of the cooling system. The WUE-site model encompasses various water usage aspects, including water utilized for space humidification, consumption at cooling towers if applicable, and any other water utilized during adiabatic processes for cooling purposes.	(Lei et al., 2023; Lei and Masanet, 2022, 2020)
Facility System Variables	Each scenario was generated utilizing the Latin Hypercube Sampling method, which samples across the uncertain range of facility system variables.	(Lei et al., 2023; Lei and Masanet, 2022, 2020)
Typical Meteorological Year (TMY) Data	For each climate zone, TMY data of the representative city (Woldekidan et al., 2020) was used for scenario generation.	Energy Plus (EnergyPlus, 2016)

1

2 Following the methodology outlined in Fig. S3.1, the 50 scenarios of facility system variables for
3 each cooling system and climate zone elucidate the difference in PUE and WUE-site between
4 efficient and inefficient data centers. These scenarios provide valuable insights into how

operational efficiency practices can impact both PUE and WUE-site, thereby enhancing our understanding of data center water use. To facilitate a meaningful comparison of water use per workload values among data centers utilizing the same cooling system and operating within a specific climate zone, we categorize facilities with PUE and WUE-site values ranked at the 5th quantiles as efficient. Conversely, those with values at the 95th quantiles are classified as inefficient.

S4: Water Consumption Factor

In this study, we focus on examining how different power sources used for electricity generation impact the water use of data centers, particularly comparing high-WCF sources like geothermal and hydroelectric power with low-WCF sources such as wind and solar PV. Since water consumption in electricity systems can vary significantly by fuel type and occurs at two main stages of electricity generation, we compiled the most recent and reliable U.S. data on consumptive water intensity. This includes upstream water use (fuel production, processing, and transportation) and direct water use at the point of generation (PoG) across different power sources, as shown in Table S4.1. Specifically, upstream water consumption covers activities related to fuel production, processing, and transportation, while water consumed at the PoG involves cooling, scrubbers, human use, and fire suppression.

Table S4.1. WCF by power source across different system boundaries.

Power Source/ System Boundary	Geother mal	Hydro	Biomass	Oil	Coal	Natural Gas	Solar PV	Wind
Upstream (Peer et al., 2019)	9.5	6.8	1.2	1.4	0.4	0.21	0.011	0.011
Point of Generation (Siddik et al., 2021)	1.59	N/A	3.0763	1.5	1.8	0.96	0.0227	0.0038
Total	11.09	6.8	4.2763	2.9	2.2	1.17	0.034	0.0148

S5: Server Electricity Use Per Workload

A comprehensive parametric analysis was conducted to investigate the influence of server utilization level, refresh cycle, the fraction of inactive servers, and efficiency on server electricity use per workload. Employing Eqs. (5) to (8), a full factorial design was implemented to represent present-day data center operational practices. This design included variables such as server utilization level ranging from 0.1 to 0.8 (Shehabi et al., 2016), server refresh cycle spanning from 1 to 8 years (Horizon Editorial, 2023; Supermicro, 2019), server management practices (reflecting either best or poor practices through the removal of needless servers, corresponding to 5% and 30% inactive servers respectively) (Kooimey and Taylor, 2017), and server power efficiency (quantified by servers ranked at the 5th and 95th quantiles in the “SPECpower_ssj 2008” database) (Standard Performance Evaluation Corporation, 2022). For more details related to the model implementations and analysis, refer to section 1 of the supporting code (SC).

Fig. S5.1 illustrates the impact of server power efficiency on server electricity use per workload by comparing low and high-efficiency server groups. The 95% confidence interval for the

difference in means of server electricity use per workload between the low and high-efficiency servers is $0.121 \text{ to } 0.156 \frac{\text{milliwatt}}{\text{ssj_ops}}$ (refer to SC: Test 1 for the conducted paired t-test), highlighting the significance of utilizing high-efficiency servers to reduce server electricity use per workload.

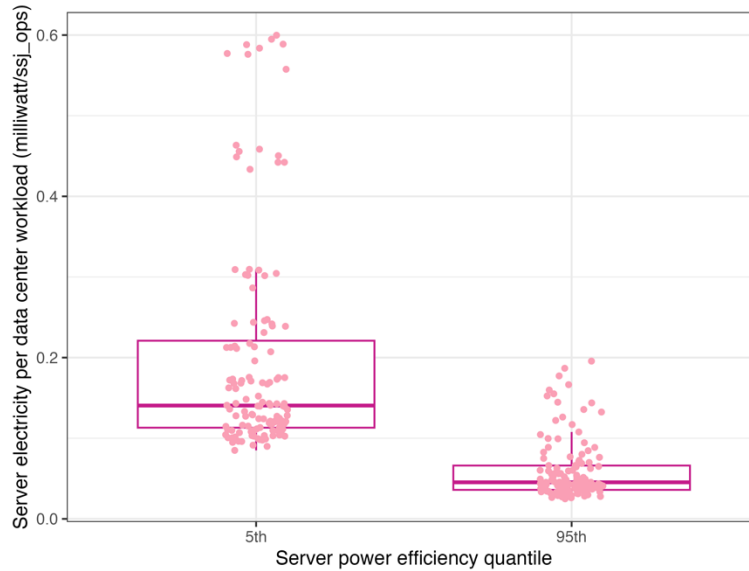


Fig. S5.1. Relationship between server electricity use per workload and server power efficiency (5th efficient ~ low server power efficiency at 5th quantile; 95th efficient ~ high server power efficiency at 95th quantile)

Fig. S5.2 depicts the relationship between server electricity use per workload and server utilization level. The solid line represents the expected server electricity use per workload at varying server utilization levels, showing a decrease as server utilization increases. The gray area represents the 95% confidence interval. The scatter points correspond to different combinations of factor values in our factorial design.

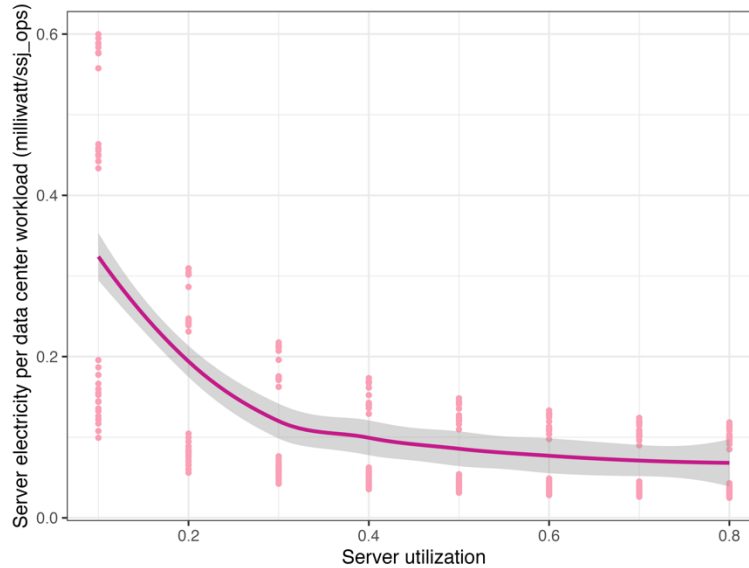


Fig. S5.2. Relationship between server electricity use per workload and server utilization level.

Fig. S5.3 shows the relationship between server electricity use per workload and server refresh cycle across varying server utilization levels, server power efficiency, and the inactive percentage of servers. Generally, a shorter refresh cycle leads to lower server electricity use per workload across all utilization levels, attributable to the higher power efficiency of newer server generations. However, the potential for server electricity savings per workload by optimizing the refresh cycle is less significant compared to improving server utilization levels or deploying high-efficiency servers, especially when servers are operating at relatively low utilization levels. Finally, the removal of needless servers within data centers can further reduce server electricity use per workload, as further demonstrated in Fig. S5.4.

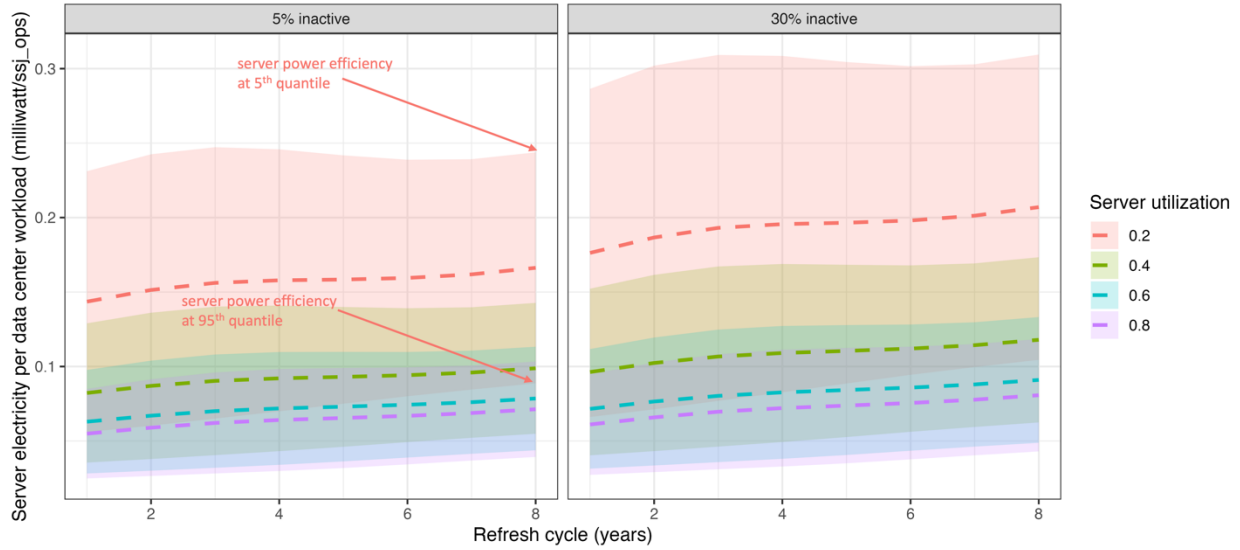


Fig. S5.3. Relationship between server electricity per workload and refresh cycle across varying utilization level, power efficiency, and inactive percentage of servers within data centers.

Fig. S5.4 illustrates the impact of inactive servers on server electricity use per workload by comparing low and high fractions of inactive servers within data centers. Removing comatose servers can reduce server electricity use per workload, as these servers consume power without performing useful computational work. The 95% confidence interval for the difference in means of server electricity use per workload between the two groups of low and high fractions of inactive servers is 0.019 to 0.031 $\frac{\text{milliwatt}}{\text{ssj_ops}}$ (refer to SC: Test 2 for the conducted paired t-test), underscoring the importance of removing comatose servers to reduce server electricity use per workload.

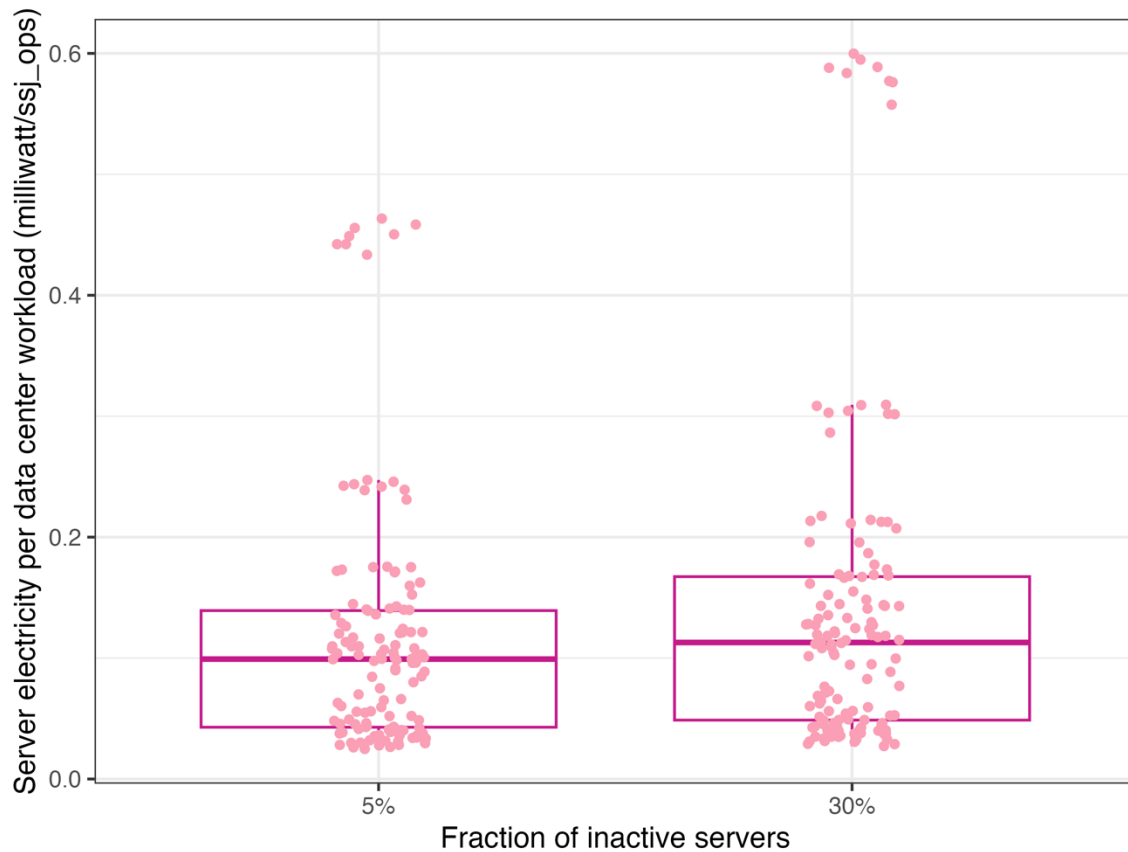


Fig. S5.4. Relationship between server electricity per workload and fraction of inactive servers.

S6: Relationship Between PUE And WUE-site values

Fig. S6.1 depicts the correlation among PUE, WUE-site, climate zones, and operational efficiency practices for data centers employing airside economizers and direct adiabatic cooling, with water-cooled chillers serving as supplementary cooling. This illustration offers a comprehensive demonstration of how a region was delineated for a specific cooling technology, as outlined in Fig. 4. The range of PUE and WUE-site values associated with a specific cooling technology stem from the interplay of climate conditions and operational efficiency of data center infrastructure. Typically, data centers situated in hot and/or humid climates exhibit higher PUE and WUE-site values, whereas those in cooler climates tend to demonstrate lower values in these metrics.

Furthermore, the difference in PUE and WUE-site values between efficient and inefficient data centers can be considerable within a particular climate context.

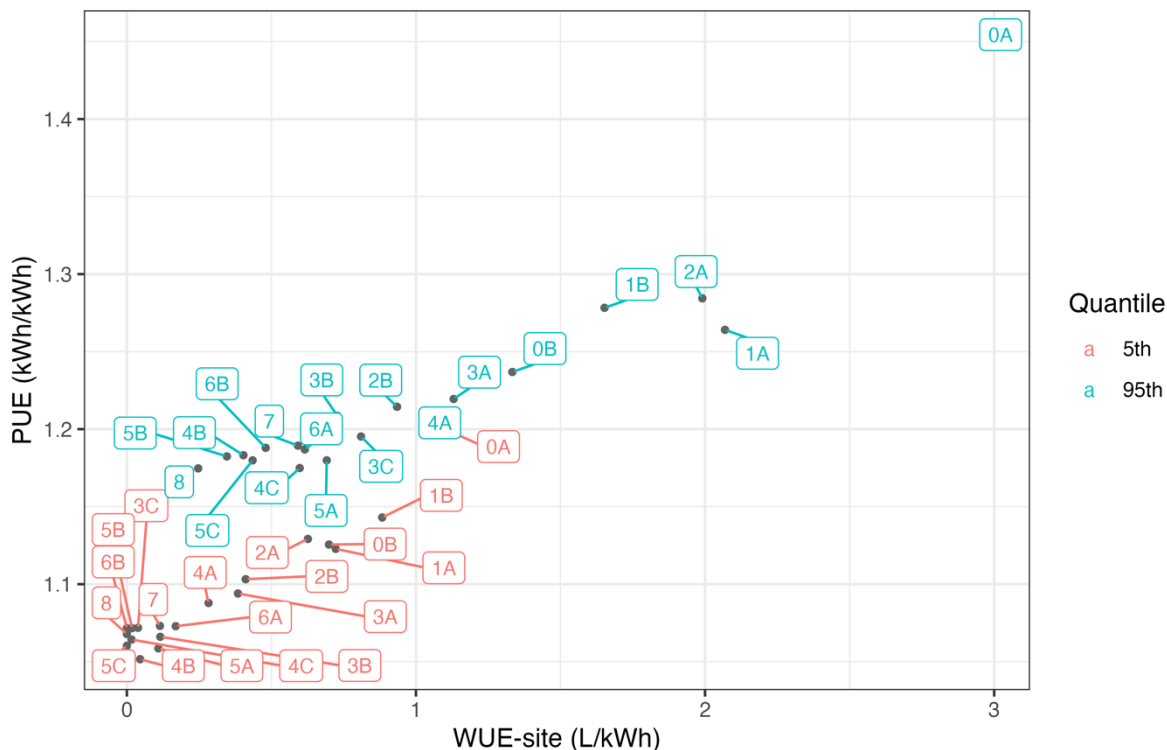


Fig. S6.1. Example of how regions are established: Relationship between PUE, WUE-site, climate zone, and operating efficiency practices (for data centers using airside economizers and direct adiabatic cooling and using water-cooled chillers as supplement cooling). 5th ~ data centers with operating efficiency scores at 5th quantile (efficient data centers, see section S3); 95th ~ data centers with operating efficiency scores at 95th quantile (inefficient data centers, see section S3).

To enhance understanding of the WUE-site values across diverse cooling technologies, we categorized these technologies into two clusters based on the magnitude of WUE-site values, as depicted in Fig. S6.2. Cluster 1 encompasses direct expansion systems, air-cooled chillers without

1 economizers, airside economizers paired with air-cooled chillers for supplementary cooling, and
2 airside economizers and direct adiabatic cooling with air-cooled chillers for supplementary
3 cooling. Cluster 2 comprises airside economizers with water-cooled chillers for supplementary
4 cooling, airside economizers and direct adiabatic cooling with water-cooled chillers for
5 supplementary cooling, water-cooled chillers without economizers, waterside economizers with
6 water-cooled chillers for supplementary cooling, IT liquid cooling with waterside economizers and
7 with water-cooled chillers for supplementary cooling, and IT liquid cooling with dry coolers with
8 adiabatic assist and with air-cooled chillers for supplementary cooling.

9 Generally, data centers employing cooling technologies in cluster 1 tend to exhibit relatively low
10 WUE-site values, due to occasional space humidification requirements. While adiabatic processes
11 are present in data centers employing airside economizers and direct adiabatic cooling with air-
12 cooled chillers for supplementary cooling, this cooling system is predominantly favored by
13 Internet Giants. Their advanced operational practices facilitate the extensive utilization of airside
14 economizers for data center cooling throughout most of the year. As a result, sporadic adiabatic
15 cooling plays a minor role, leading to minimal water consumption in these facilities. On the other
16 hand, data centers utilizing cooling technologies in cluster 2 generally demonstrate higher WUE-
17 site values. This is primarily attributed to the utilization or potential utilization of cooling towers
18 or the wet-mode operation of dry coolers throughout the year. In these cases, substantial volumes
19 of water are evaporated to effectively dissipate heat from the data centers, contributing to elevated
20 water consumption levels. Within this cluster, data centers utilizing IT liquid cooling with dry
21 coolers featuring adiabatic assist and employing air-cooled chillers for supplementary cooling
22 showcase the lowest WUE-site values. This is primarily due to the elevated facility water
23 temperatures characteristic of liquid-cooled IT systems. These elevated temperatures facilitate the

dry-mode operation of the dry coolers, effectively fulfilling data center cooling requirements for a significant portion of the year. Consequently, the more water-intensive wet-mode operations are only necessitated for a minor percentage of the year.

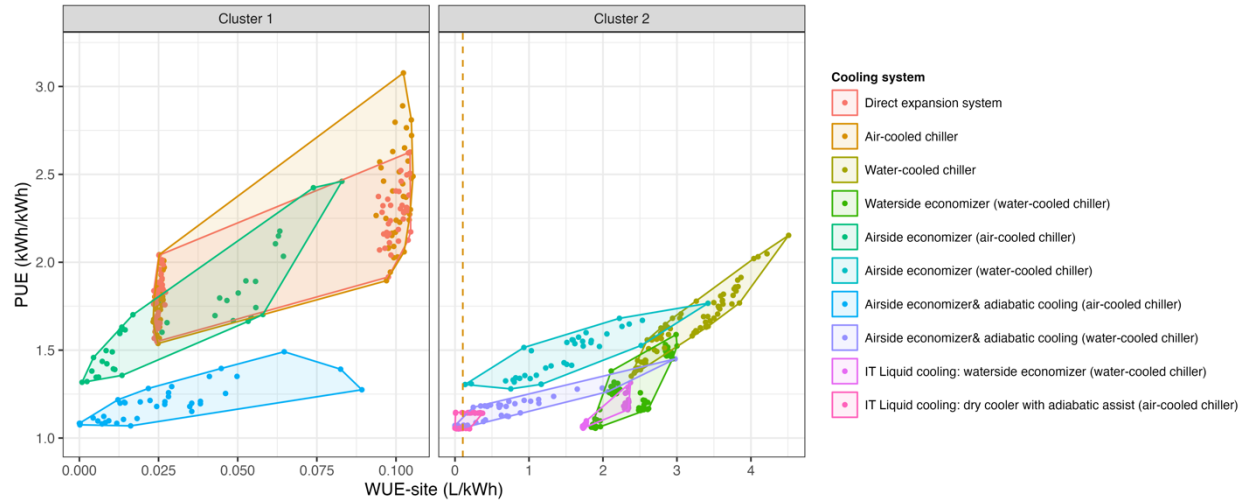


Fig. S6.2. Relationship between PUE, WUE-site, and cooling technology

Note: x-axis values are linearly spaced.

In this study, we investigated two major liquid cooling systems (ASHRAE, 2014; Evans, 2004): (1) IT liquid cooling with dry coolers assisted by adiabatic cooling and air-cooled chillers as supplemental cooling, and (2) IT liquid cooling with waterside economizers and water-cooled chillers as supplemental cooling. Within these two systems, three liquid-cooled IT equipment scenarios were considered (ASHRAE, 2014): (a) rear door heat exchanger, (b) cold plate (direct-to-chip cooling), and (c) immersion cooling. Table S6.1 summarizes the key assumptions used in the PUE and WUE-site simulations for different IT equipment liquid cooling scenarios, while Fig. S6.3 presents the corresponding PUE and WUE-site simulation results for each liquid cooling type and scenario.

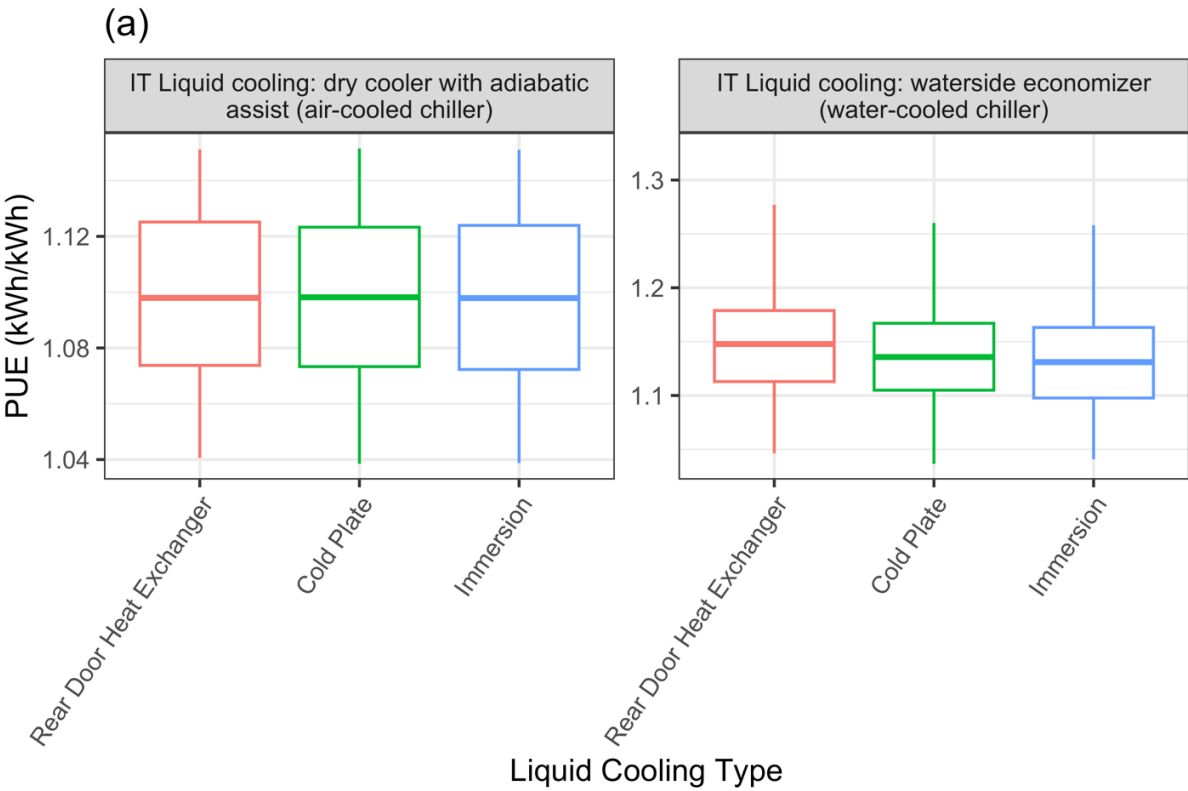
Unlike the significant variations in PUE and WUE-site values across different cooling technologies shown in Fig. 4, the differences in PUE and WUE-site values across the various liquid-cooled IT equipment scenarios are relatively small. The yearly average PUE and WUE-site values are highest for rear door heat exchanger cooling, lowest for immersion cooling, with cold plate (direct-to-chip cooling) in between. However, this trend is difficult to discern in Fig. S6.3 due to two major reasons. First, the allowable facility water supply temperatures for different liquid-cooled IT equipment scenarios are relatively broad (Table S6.1), with significant overlap. This allows free cooling and adiabatic cooling to meet data center requirements without needing supplemental mechanical cooling for most of the year, even in warmer or more humid climate zones. In free or adiabatic cooling mode, the PUE and WUE-site values are nearly identical across these systems when similar efficiency practices are used (e.g., UPS efficiency, pumping system efficiency). Second, PUE values in liquid cooling systems are already quite low, while WUE-site values are largely influenced by the choice of heat rejection components (e.g., cooling towers, dry coolers) within the cooling system. In these scenarios, even minor reductions in PUE and WUE-site require substantial system-wide optimization, rather than merely adjusting IT equipment cooling strategies. This highlights the growing need to develop new metrics for measuring data center infrastructure efficiency in the future.

Table S6.1. Facility water supply temperature for different Liquid Cooling Scenario (ASHRAE, 2014; ASHRAE Technical Committee 9.9, 2011).

Liquid-cooled IT Equipment Scenario	Facility Water Supply Temperature (°C)

Rear Door Heat Exchanger	2-32
Cold Plate (Direct to Chip)	2-40
Immersion Cooling	2-45

1



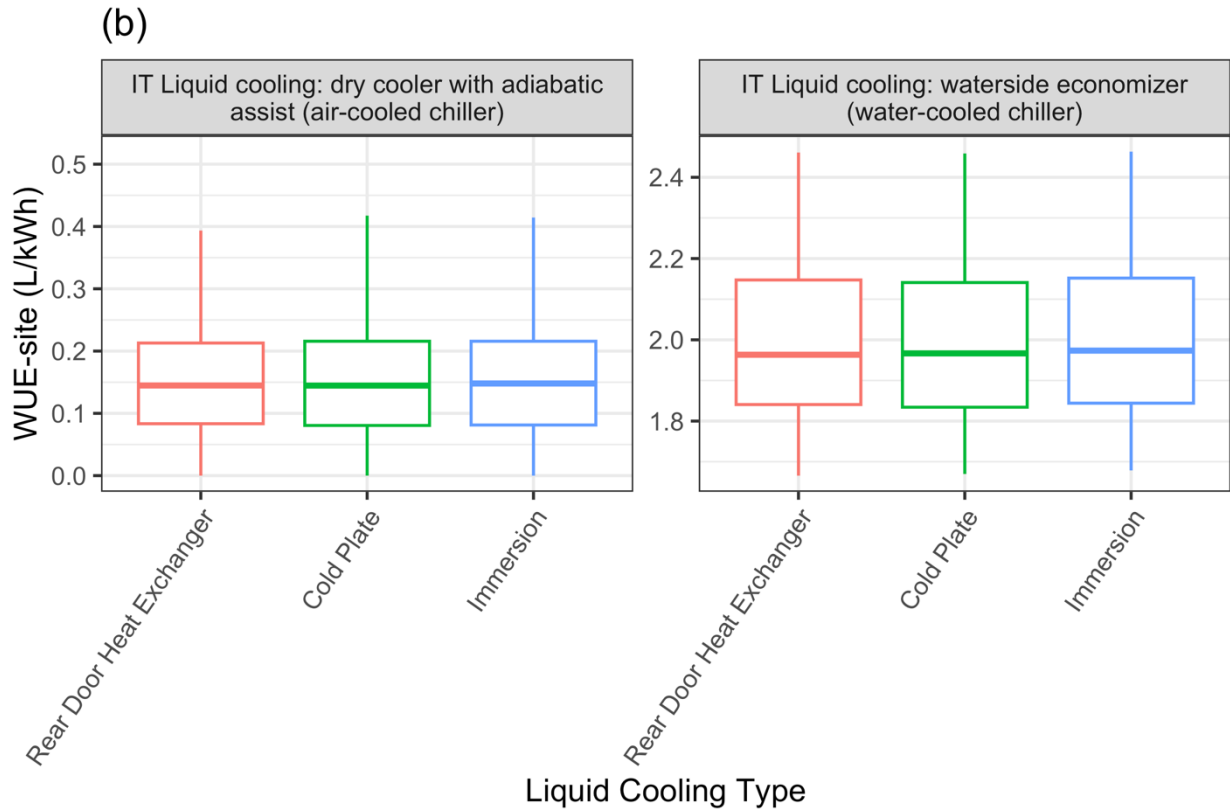


Fig. S6.3. PUE and WUE values for different liquid cooling system and IT equipment cooling scenarios. Note: the y-axis scale differs from left to right.

1

2 **S7: Facility-level electricity and water use per workload**

3 Like PUE and WUE-site, facility-level electricity and water use per workload also exhibit a
 4 delicate relationship due to the multiplicative effect of server electricity use per workload (Eq. (2)
 5 and (3)). For a clear illustration, we visualize the relationship between facility electricity and water
 6 use per workload for data centers using airside economizers and adiabatic cooling and using water-
 7 cooled chillers as auxiliary cooling systems, which result in a region in Fig. S7.1. This region is
 8 bounded left to right by the product of server workload efficiency and possible WUE-site values,
 9 and top to bottom by the product of server workload efficiency and possible PUE values. This

clearly indicates that the lowest data center facility-level electricity and water use per workload is associated with the highest server workload efficiency.

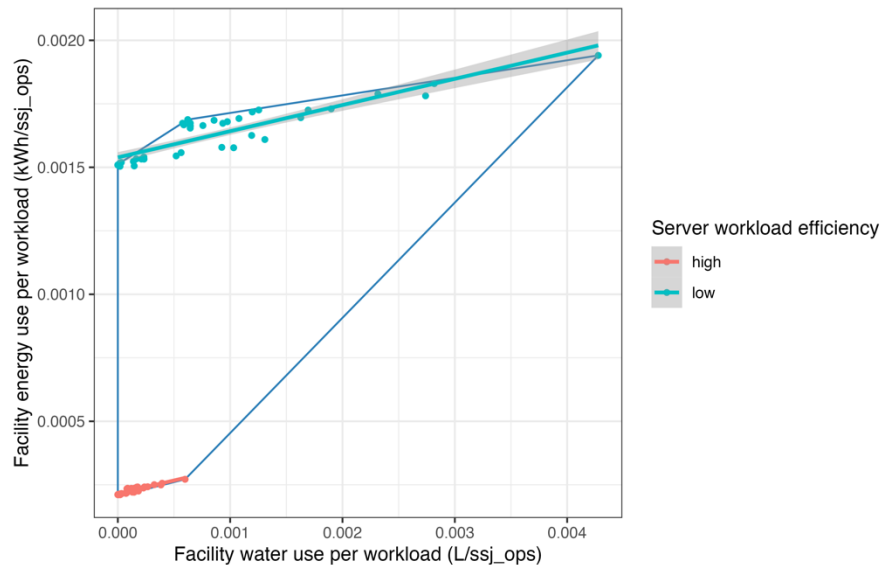


Fig. S7.1. Relationship between facility electricity and water use per workload in data centers with airside economizer and adiabatic cooling (water-cooled chiller) systems (Low and high server workload efficiency represents minimum and maximum server electricity per workload in Fig. 2).

S8: WUE-source

Based on the PUE and WUE-site values from Fig. 4 and the WCFs from Fig. 5(b), Fig. S8.1 illustrates the relationship between WUE-source, WUE-site, and WCF. Two examples are used: data centers with airside economizers and adiabatic cooling (water-cooled chiller), and data centers with direct expansion systems. As discussed in Section 3.5, WUE-source is linearly related to both

- 1 WCF and PUE. The highest and lowest WUE-source values correspond to the highest and lowest
- 2 WCFs, as demonstrated by the lines for geothermal and wind in Fig. S8.1.

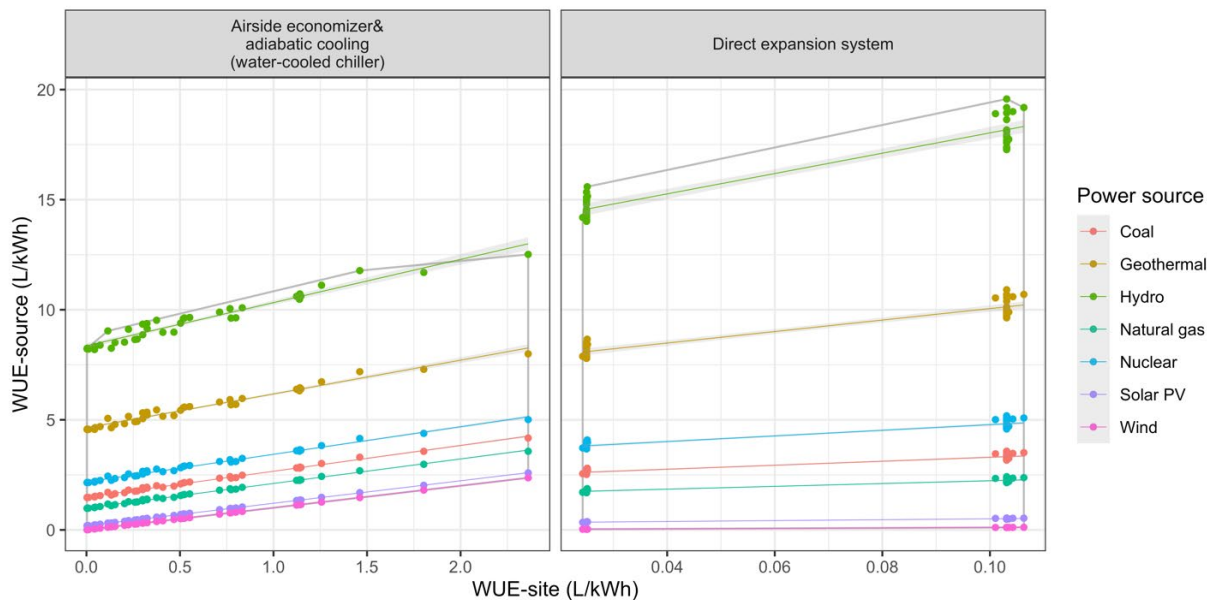


Fig. S8.1. Relationship between WUE-source, WUE-site, and WCF in data centers with airside economizer and adiabatic cooling (water-cooled chiller) or direct expansion systems (Note: the lines for Solar PV and Wind are very close together due to the y-axis scale).

- 3
- 4 **S9: Overall Water Use Per Workload**
- 5 Fig. S9.1 illustrates the relationship between facility-level and overall water use per workload for
- 6 data centers with various cooling systems and power sources, including hydro, biomass, oil, coal,
- 7 natural gas, and solar PV, using WCF data introduced in Section S4. As discussed in Section 3.6,
- 8 data centers powered by low-WCF sources generally have much lower overall water use per
- 9 workload compared to those using high-WCF power. However, utilizing a low-WCF power source
- 10 does not guarantee low overall water use per workload, especially when the data center has high

- 1 PUE, WUE-site, and low server workload efficiency, as demonstrated by the upper right region of
- 2 data centers powered by solar PV in Fig. S9.1 below.

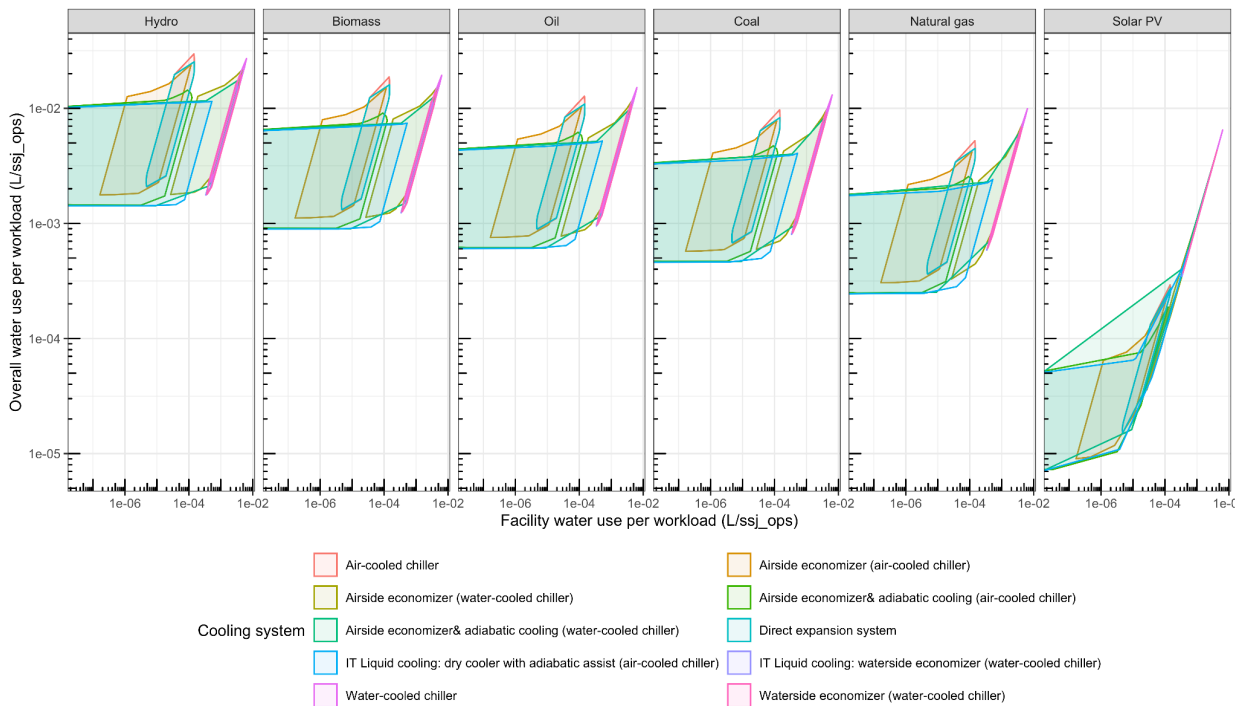


Fig. S9.1. Relationship between data center facility-level and overall water use per workload by cooling technology for data centers that rely on hydro, biomass, oil, coal, natural gas, and solar PV based electricity.

3

4 **S10: Data Source Summary**

5 Table S10. A summary of the inputs and data sources used in this analysis.

Component	Variable	Reference Sections	Data Sources
Direct server electricity use per workload	Server utilization level	Section 2.2, S2, S5	(Shehabi et al., 2016)

	Vintage distribution of installed server stock	Section S1, S5	(Horizon Editorial, 2023; Supermicro, 2019)
	Server efficiency	Section S2, S5	(Standard Performance Evaluation Corporation, 2022)
	Fraction of inactive servers	Section 2.2, S5	(Koomey and Taylor, 2017)
PUE and WUE-site	PUE	Section 2.3, S3, S6	(ASHRAE, 2014, 2013; ASHRAE Technical Committee 9.9, 2011; EnergyPlus, 2016; Lei, 2022; Lei et al., 2023; Lei and Masanet, 2022, 2020)
	WUE-site	Section 2.3, S3, S6	(ASHRAE, 2014, 2013; ASHRAE Technical Committee 9.9, 2011; EnergyPlus, 2016; Lei, 2022; Lei et al., 2023; Lei and Masanet, 2022, 2020)
Electric power water consumption factor	WCF (Upstream)	Section 2.4, S4	(Peer et al., 2019)
	WCF (PoG)	Section 2.4, S4	(Siddik et al., 2021)

1

2 **References**

- 3 ASHRAE, 2014. Liquid Cooling Guidelines for Datacom Equipment Centers, 2nd Edition,
4 ASHRAE Datacom Series. Atlanta.
- 5 ASHRAE, 2013. Climatic data for building design standards. ASHRAE Stand.
- 6 ASHRAE Technical Committee 9.9, 2011. Thermal Guidelines for Data Processing
7 Environments, American Society of Heating, Refrigerating, and Air-Conditioning
8 Engineers Inc.
- 9 Bowman, A.A.A., 2018. Package “sm.”
- 10 EnergyPlus, 2016. Weather Data – EnergyPlus [WWW Document]. US Department Energy.
11 URL <https://energyplus.net/weather>
- 12 Evans, T., 2004. The different types of air conditioning equipment for IT environments. APC
13 White Pap. 24.
- 14 Horizon Editorial, 2023. Navigating Hardware Refresh Cycles In The Data Center [WWW

Document]. URL <https://horizontechnology.com/news/data-center-hardware-refresh-cycles/> (accessed 4.22.24).

Hurvich, C.M., Simonoff, J.S., Tsai, C.L., 1998. Smoothing parameter selection in nonparametric regression using an improved Akaike information criterion. *J. R. Stat. Soc. Ser. B Stat. Methodol.* 60, 271–293. <https://doi.org/10.1111/1467-9868.00125>

Kafadar, K., Bowman, A.W., Azzalini, A., 1999. Applied Smoothing Techniques for Data Analysis: The Kernel Approach with S-PLUS Illustrations. *J. Am. Stat. Assoc.* <https://doi.org/10.2307/2670015>

Koomey, J., Taylor, J., 2017. ZOMBIE/COMATOSE SERVERS REDUX.

Lei, N., 2022. A Hybrid Physics-Based and Data-Driven Modeling Framework for Energy and Water Use Analysis of Data Centers with Spatio-Temporal Resolution. Northwestern University, Evanston, IL, USA.

Lei, N., Lu, J., Cheng, Z., Cao, Z., Shehabi, A., Masanet, E., 2023. Geospatial assessment of water footprints for hyperscale data centers in the United States. *J. Phys. Conf. Ser.* 2600, 172003. <https://doi.org/10.1088/1742-6596/2600/17/172003>

Lei, N., Masanet, E., 2022. Climate- and technology-specific PUE and WUE estimations for U.S. data centers using a hybrid statistical and thermodynamics-based approach. *Resour. Conserv. Recycl.* 182, 106323. <https://doi.org/10.1016/J.RESCONREC.2022.106323>

Lei, N., Masanet, E., 2020. Statistical analysis for predicting location-specific data center PUE and its improvement potential. *Energy* 201, 117556. <https://doi.org/10.1016/j.energy.2020.117556>

Peer, R.A.M., Grubert, E., Sanders, K.T., 2019. A regional assessment of the water embedded in the US electricity system. *Environ. Res. Lett.* <https://doi.org/10.1088/1748-9326/ab2daa>

Shehabi, A., Smith, S.J., Sartor, D.A., Brown, R.E., Herrlin, M., Koomey, J.G., Masanet, E.R., Horner, N., Azevedo, I.L., Lintner, W., 2016. United States Data Center Energy Usage Report. Berkeley Lab 65.

Siddik, M.A.B., Shehabi, A., Marston, L.T., 2021. The environmental footprint of data centers in the United States. *Environ. Res. Lett.* <https://doi.org/10.1088/1748-9326/abfba1>

Standard Performance Evaluation Corporation, 2022. SPECpower_ssj® 2008 [WWW Document]. URL https://www.spec.org/power_ssj2008/ (accessed 6.22.22).

Supermicro, 2019. Data Centers & The Environment: 2019 Report on the State of the Green Data Center.

Woldekidan, K.F., Torcellini, P.A., Langner, M., Pflaum, N., 2020. Moving Zero Energy Office Buildings to the Mainstream: Establishing Design Guidelines and Energy Targets: Preprint.